

Single-cell RNA sequencing of the holothurian regenerating intestine reveals the pluripotency of the coelomic epithelium

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Abstract

In holothurians, the regenerative process following evisceration involves the development of a “rudiment” or “anlage” at the injured end of the mesentery. This regenerating anlage plays a pivotal role in the formation of a new intestine. Despite its significance, our understanding of the molecular characteristics inherent to the constituent cells of this structure has remained limited. To address this gap, we employed state-of-the-art scRNA-seq and HCR-FISH analyses to discern the distinct cellular populations associated with the regeneration anlage. Through this approach, we successfully identified thirteen distinct cell clusters. Among these, two clusters exhibit characteristics consistent with putative mesenchymal cells, while another four show features akin to coelomocyte cell populations. The remaining seven cell clusters collectively form a large group encompassing the coelomic epithelium of the regenerating anlage and mesentery. Within this large group of clusters, we recognized previously documented cell populations such as muscle precursors, neuroepithelial cells and actively proliferating cells. Strikingly, our analysis provides data for identifying at least four other cellular populations that we define as the precursor cells of the growing anlage. Consequently, our findings strengthen the hypothesis that the coelomic epithelium of the anlage is a pluripotent tissue that gives rise to diverse cell types of the regenerating intestinal organ. Moreover, our results provide the initial view into the transcriptomic analysis of cell populations responsible for the amazing regenerative capabilities of echinoderms.

eLife assessment

This study presents a dataset obtained through a single cell RNA-Sequencing of sea cucumber regenerating intestine 9 days post evisceration. The data were collected and analyzed using standard single cells analysis from $n=2$ adult sea cucumbers captured from the wild, which represents a **useful** resource for future studies. Although cell type validation is attempted, it is performed on samples from the same 2 animals (and not independent samples), rendering the validation **incomplete**. Further, the RNA localization images provided in the paper could benefit from improved spatial context, and many strong statements in the discussion should be better justified and supported by the presented data. With the validation part strengthened, this paper would be of interest to development and regeneration fields.

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Introduction

Animals exhibit a remarkable diversity in their responses to injury, ranging from basic wound healing to the complete regeneration of lost structures. At one end of the spectrum are species that heal wounds without regenerating the missing part, while at the other end are those capable of recreating structures identical to the original. Despite these differences, all animals possess some level of regenerative ability. This capacity for regeneration can vary not only between different species but also within the same species, depending on the tissue or organ involved.

For decades, scientists have tried to understand these vast differences in regeneration capacities across the animal kingdom. For this they have focused on those species that show amazing regenerative abilities (Tanaka and Reddien, 2011 [↗](#)), including coelenterates (hydra) (Vogg et al., 2019 [↗](#)), flatworms (planaria) (Reddien, 2018 [↗](#)), fish (zebrafish) (Gemberling et al., 2013 [↗](#)) and amphibians (axolotl) (Roy and Gatién, 2008 [↗](#)) among others. These studies have uncovered various mechanisms that regeneration-competent species exhibit to regenerate tissues, entire organs, body parts, and in some cases complete bodies. Key findings include the discovery of essential processes, such as the formation of a blastema – a mass of proliferating cells that plays a crucial role in regenerating the lost structure (Min and Whited, 2023 [↗](#); Poleo et al., 2001 [↗](#); Santos-Ruiz et al., 2002 [↗](#); Seifert and Muneoka, 2018 [↗](#); Stocum, 2004 [↗](#); Zenjari et al., 1996 [↗](#)).

Among deuterostomes, echinoderms are considered prime exponents of regenerative capability (Candia-Carnevali et al., 2024 [↗](#)). Within this group, holothurians, commonly known as sea cucumbers, exhibit an extraordinary form of regeneration. They can regenerate their internal organs following evisceration, a process in which they expel their viscera in response to stress or predation (Byrne, 2023 [↗](#)). This extraordinary ability makes them a valuable model for studying regeneration in complex organisms. The regeneration of the digestive system in holothurians, in particular, has garnered significant interest (Dolmatov, 2021 [↗](#); Mashanov et al., 2014 [↗](#); Medina-Feliciano and García-Arrarás, 2021 [↗](#); Quispe-Parra et al., 2021 [↗](#); Su et al., 2022 [↗](#)). Upon evisceration, the holothurian intestine, which constitutes nearly their entire digestive tract, begins to regenerate from the mesentery, a supportive tissue layer where the original intestine was attached (García-Arrarás et al., 1998 [↗](#)). A thickening at the injured end of the mesentery, known as an “anlage” or “rudiment,” initiates this process (García-Arrarás et al., 1998 [↗](#)). This structure is

analogous to a blastema but differs in that the cell proliferation mainly occurs in the surrounding epithelium, rather than in the mesenchymal cells, as observed in classical blastemas (Carlson, 2007 [↗](#); García-Arrarás et al., 1998 [↗](#)).

Histologically, the holothurian intestinal anlage forms from dedifferentiated cells within the mesentery, which revert to a more stem-cell-like state before proliferating and migrating to form a new intestinal structure (Candelaria et al., 2006 [↗](#); Mashanov et al., 2005 [↗](#)). This dedifferentiation process is crucial for regeneration, involving a spatial and temporal gradient starting at the injury site and extending along the mesenteric border (García-Arrarás et al., 2011 [↗](#)). The new epithelial layer that forms around the anlage is distinct from the original mesothelium, shows significant morphological and molecular changes compared to the mesenteric tissue (Candelaria et al., 2006 [↗](#); García-Arrarás et al., 1998 [↗](#); Mashanov et al., 2005 [↗](#)). Further examination of holothurian regeneration reveals that most cellular division occurs within the anlage's coelomic epithelium (García-Arrarás et al., 2011 [↗](#), 1998 [↗](#)). These proliferating cells are hypothesized to differentiate into various cell types, including myocytes and neurons, as well as mesenchymal cells, through an epithelial to mesenchymal transition (EMT) (García-Arrarás et al., 2011 [↗](#), 1998 [↗](#)). The gene expression profiles during the formation and growth of the anlage suggest extensive reprogramming, leading to a more plastic cell phenotype (Ortiz-Pineda et al., 2009 [↗](#); Quispe-Parra et al., 2021 [↗](#); Rojas-Cartagena et al., 2007 [↗](#)).

The process of intestinal regeneration in holothurians raises fundamental questions of regenerative phenomena particularly concerning the identity, origin, and fate of progenitor cells, involved in the process (Alvarado and Tsonis, 2006 [↗](#); Candia-Carnevali et al., 2024 [↗](#)). In this context, the role of the anlage, and specially the mesothelium (also named celothelium) in echinoderms, deserved particular attention (Candia-Carnevali et al., 2024 [↗](#); Smiley, 1994 [↗](#)). This tissue, mainly composed of coelomic epithelia and myocytes, exhibit significant morphological and gene expression changes that are associated with the dedifferentiation process. These dedifferentiated cells form the coelomic epithelium of the anlage and appear to be the principal source of cells for the new intestine. Despite these findings, little is known about the cell composition and dynamics of the anlage nor of its coelomic epithelium.

Single-cell omic studies offer tremendous promise to dissect the cellular contributions of the holothurian intestinal anlage and to identify the specific cells involved in generating a new intestine. Single-cell RNA sequencing (scRNA-seq) a powerful tool for dissecting cellular composition and dynamics has been used in related species to explore regenerating or developing tissues. In mice, for instance, scRNA-seq has been employed to investigate the response to injury. However, these studies have primarily focused on tissue healing rather than the full regenerative process (Ayyaz et al., 2019 [↗](#); Beumer and Clevers, 2021 [↗](#); Capdevila et al., 2021 [↗](#); Parigi et al., 2022 [↗](#)). Beyond healing, scRNA-seq has been pivotal in studying regenerating systems such as the axolotl limb blastema and planarian regeneration providing a detailed insights into the cellular processes at play (Gerber et al., 2018 [↗](#); King et al., 2024 [↗](#); Leigh et al., 2018 [↗](#); Rodgers et al., 2020 [↗](#)). Furthermore, considering the close relationship between regeneration and embryogenesis, scRNA-seq, has also been used to understand the cellular events in developing echinoderm embryos (Cocurullo et al., 2023 [↗](#); Paganos et al., 2022a [↗](#), 2022b [↗](#), 2021 [↗](#); Satoh et al., 2022 [↗](#); Tominaga et al., 2023 [↗](#)). These applications highlight the groundbreaking role of scRNA-seq in advancing our knowledge of the cellular mechanisms in both regenerative and developmental contexts.

In this study, we employ scRNA-seq to analyze the regenerating intestinal anlage of the sea cucumber *Holothuria glaberrima*, aiming to delineate its constituent cellular populations (**Fig 1A** [↗](#)). We corroborate our findings using hybridization chain reaction fluorescent in situ hybridization (HCR-FISH) to verify the presence and location of specific cell types (Choi et al., 2018 [↗](#)). The results provides transcriptomic evidence on the pluripotency of the anlage coelomic epithelia and the first description of the holothurian regenerating precursor cells. This research

not only advances our understanding of the unique regenerative capabilities of holothurians but also contributes to the broader field of regenerative biology, highlighting the diverse strategies employed by different organisms to restore lost tissues.

Results

Previous work from our laboratory has shown that the rudiment or anlage that forms at the tip of the mesentery plays a pivotal role in the formation of the new intestine (García-Arrarás et al., 2019 [↗](#)). This transient mass of cells is thought to give rise to most intestinal cell types, the sole exception being the luminal cells. Therefore, to maximize the characterization of the cells in the regenerative anlage, we chose to perform scRNA-seq in the tissues of 9-days post evisceration (dpe) regenerating animals (García-Arrarás et al., 2011 [↗](#), 1998 [↗](#)). At this stage a well-formed anlage consists of epithelial cells surrounding a large area of connective tissue populated with mesenchymal cells (Fig 1A [↗](#)). More importantly, different cell populations undergoing proliferation or differentiation can be found at this stage. Since some of the cellular processes during regeneration occur in a spatio-temporal gradient along the length of the mesentery, we separately surveyed the anlage and the mesentery tissues accordingly. Thus, for each animal, the anlage was separated from the mesentery, and both tissues were processed independently, for a total of four scRNA-seq runs.

Tissues were treated with 0.05% trypsin for 15 min and cells were dissociated manually. The dissociated cells were counted and used for the scRNA-seq analyses. Additionally, samples from the cell dissociation were fixed in poly-lysine treated slides and analyzed with immuno- and cyto-chemical techniques. Therefore, these cells correspond to the same batch whose transcriptomes were sequenced.

Cell Heterogeneity: Immuno- and cyto-chemical analyses

The strength of the scRNA-seq data depends mainly on the dissociation and isolation of the cell populations from the dissected tissue. Since our focus was on the cells of the regenerating anlage, we devised a dissociation protocol that favored the isolation of the cells within this structure. To determine, at least partially, the cell types in our original dissociation, we performed immuno- and cyto-chemical analyses on the enzyme-dissociated cell suspension that was used for the scRNA-seq. The labeling obtained for each marker is shown in S1 Fig [↗](#).

We found that, in the dissociated anlage, 5% of the cells were labeled with a spherulocyte (immune cell) marker, 2-7% with phalloidin (a muscle marker), 40% with a mesenchymal marker (KL14-antibody), and a large number of cells (32-60%) with a mesothelial marker (MESO-antibody). Similar populations were found in the mesentery, although in this tissue, 5-10% of cells expressed the neuronal markers, heptapeptide GFSKLYFamide (Díaz-Miranda et al., 1995 [↗](#)) and RN1 (Díaz-Balzac et al., 2007 [↗](#)). This analysis suggested that most of the cells originate from the regenerate coelomic epithelium.

Different populations were also observed with three different tubulin antibodies (S2 Fig [↗](#)). In the 9-dpe anlage, anti-acetylated alpha tubulin labeled about 7% of the cells, anti-beta tubulin labeled about 70% of the cells, while an anti-alpha tubulin labeled about 80% of the cells. Except for the acetylated alpha tubulin, which labeled about 1% of the cells dissociated from the mesentery versus 7% of those from the anlage, anti-alpha and beta labeling percentages were similar in both mesentery and anlage dissociated cells.

Finally, a cell population labeled with fluorescently-labeled phalloidin accounted for about 7% of the cells in both mesentery and anlage. These cells, however, did not correspond to the elongated muscle cells of the mesentery. Instead, they were rounded cells with labeling found in the

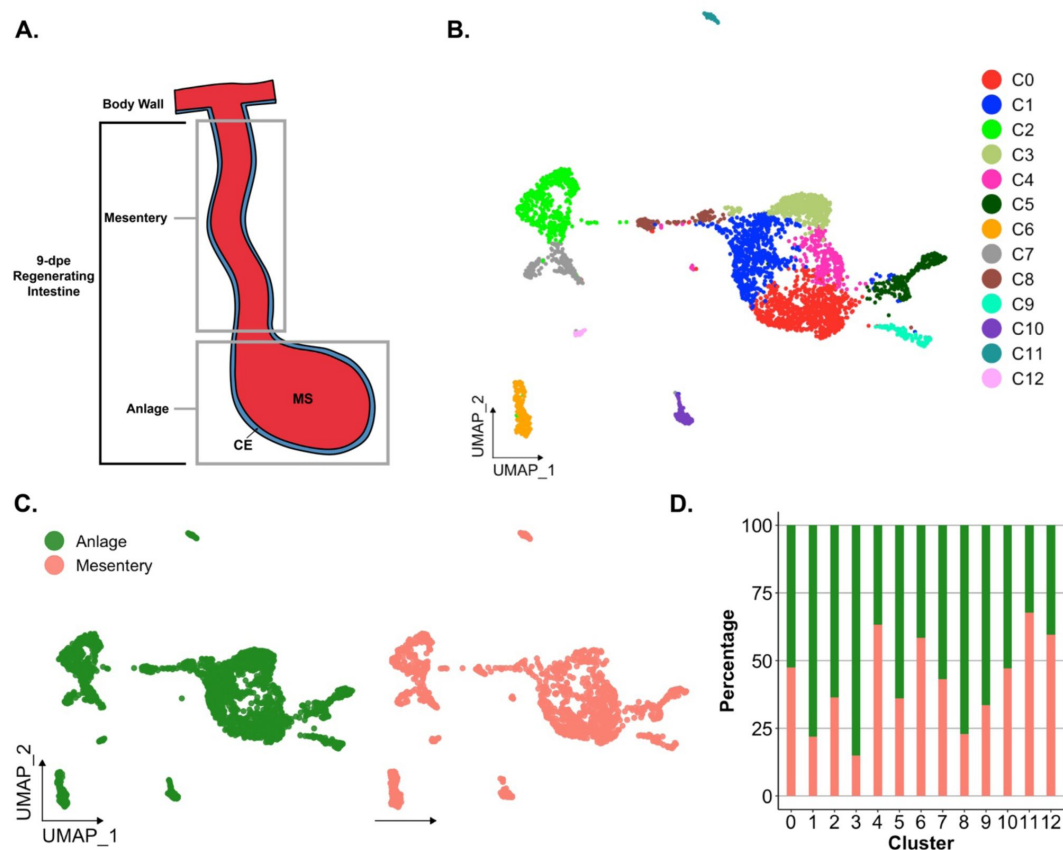


Fig 1.

Overview of single cell RNA sequencing of regenerating intestinal tissue of *H. glaberrima*.

(A) Diagram of 9-day regenerating intestine depicting mesentery and anlage components (B) UMAP plot of population identities determined through unsupervised clustering of 9-day regenerating mesentery and anlage tissues. (C) Representation of cells per tissue of both samples. (D) UMAP projections of cluster cells separated by tissue of origin. (E) Percentage of cells per cluster based on their tissue of origin.

cytoplasm surrounding one side of the nuclei.

Two cell populations from the mesentery are not represented or are greatly underrepresented in the scRNA-seq. Firstly, the muscle cells of the mesentery, due to their non-dissociation by the protocol used, were absent in the dissociated cell suspension. Their elongated morphology would have further complicated their passage via cell separation system for sequencing. Secondly, the majority of neurons from the neuronal network associated with the mesentery could not be dissociated (Nieves-Ríos et al., 2020 [↗](#)). We did, however, observe structures that resembled a tangled mass of cells immunoreactive to some of our neuronal markers. This suggests that the mesentery nervous component, being unable to be isolated as single cells, was not sequenced.

In summary, the immuno- and cytochemical results show that the dissociated cell populations sequenced correspond to cellular phenotypes that have been previously described within the regenerating anlage. The abundance of these cells in the sequenced samples corresponds to the ease of their dissociation by trypsin. Thus, dedifferentiated cells of the mesentery and anlage epithelium (which are loosely connected to each other) and those of the connective tissue are probably over-represented compared to differentiated cell types.

Cell populations defined by scRNA-seq

Analysis of scRNA-seq data resulted in a total of 3,844 cells, with 2,392 originating from the two anlage samples and 1,452 from the two mesentery samples. Upon dataset integration and graph-based clustering, we identified 13 clusters, each thought to represent singular cell types or cell states in the regenerating intestine (**Fig 1A-B** [↗](#)). The percentage of cells that form each cluster differs from cluster to cluster, ranging from 21% (cluster 0) to 1% (cluster 12) of the total cells. Nonetheless, each cluster consists of cells from both the mesentery and anlage samples (**Fig. 1C** [↗](#) & **D** [↗](#)). The number of clusters did not change dramatically under various parameters (resolution, dimensionality, and number of variable features), constantly around 12-15 clusters. Moreover, except for C3 and C4, the clusters are supported by the clustering significance analysis performed with scSHC (Grabski et al., 2023 [↗](#)), a model-based hypothesis testing method for scRNA-seq that evaluates the probability of each individual cluster being unique (**S3 Fig** [↗](#)). The uniqueness of C3 and C4 is suggested by additional analyses as will be addressed below.

Each identified cluster exhibits a distinctive gene expression profile relative to cells in other clusters (**Fig 2** [↗](#)). The top expressed gene serves as a marker for each cell population to highlight the uniqueness of each cluster. In **Fig 2** [↗](#), we show the relative expression of the top gene for each cluster based on two factors: (1) difference in percentage of representation and (2) $\log_2$ fold-change ($\log_2$ FC) against all other clusters. Interestingly, each cluster shows dramatic differential expression values ($> 2 \log_2$ FC) and differences in representation percentages over 50%, except for C0 through C3, with differences in representation around 30%.

Prior to characterizing each of the 13 cell clusters, we sought to understand what, in a broad view, appears to be a segregation of ~90% of the cells into two distinct supra-clusters. One of them encompassing 7 clusters (C0, C1, C3, C4, C5, C8, C9) that corresponded to 69.6% of all cells and the other encompassing two clusters (C2 and C7) that corresponded to 19.1% of cells. The remaining 11.3% of cells were distributed in 4 distinct isolated clusters (C6, C10, C11, and C12). As stated earlier, all clusters have representation from mesentery and anlage tissues (**Fig 1C** [↗](#)-**D** [↗](#)), thus excluding the possibility that the two supra-clusters represented mesentery versus anlage cells.

Mesenchymal versus Epithelial Clusters

These two supra-clusters are of interest as they appear to represent the two main cell types found in the regenerating intestine: coelomic epithelial cells and mesenchymal cells (**Fig 1B** [↗](#)). Understanding the cell types that correspond to each cluster is essential as cellular processes crucial for regeneration are thought to be localized to the coelomic epithelia of the regenerating

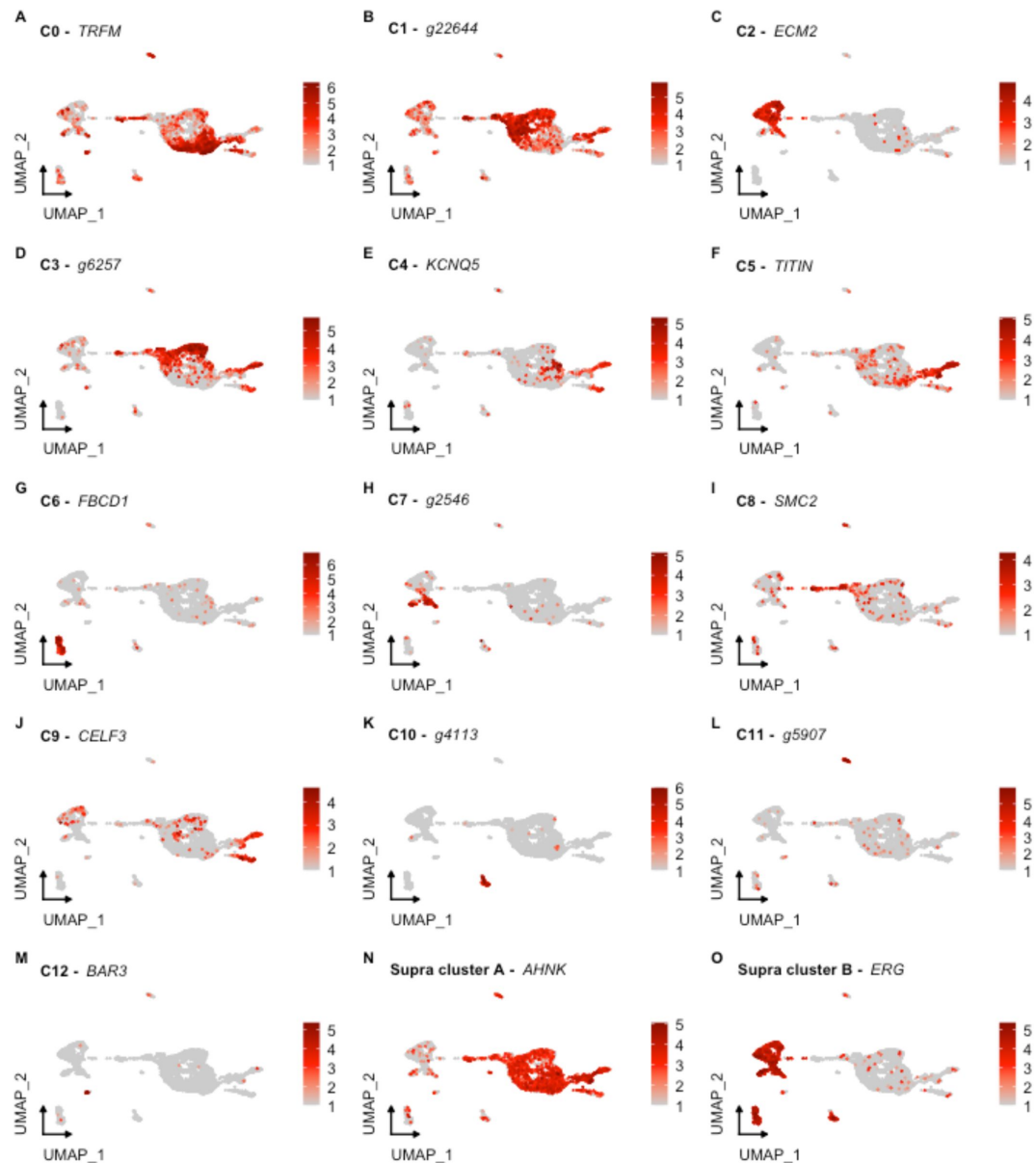


Fig 2.

UMAP visualization of clusters highlighting the expression of their top genes.

Each gene corresponds to the top gene of each independent cluster based on the percentage of representation to other clusters.

anlage. Comparison between these two supra-clusters showed distinct expression profiles that allowed us to characterize their cell types. For instance, the supra-cluster composed of C2 and C7, showed *ERG* (transcriptional regulator ETS-related gene) as the top expressed gene, an oncogene that is related to embryonic development, differentiation, angiogenesis, and apoptosis (Dhordain et al., 1995 [↗](#); Iwamoto et al., 2001 [↗](#); Vlaeminck-Guillem et al., 2000 [↗](#)) (**Fig 2** [↗](#)). More importantly, the expression of *ERG* has been associated with mesenchymal identity in other echinoderms (Meyer et al., 2023 [↗](#); Tominaga et al., 2023 [↗](#)). In these studies, *ERG* has been reported as the marker gene of mesenchymal cells of sea urchin and sea star larva by scRNA-seq analyses and the localization of *ERG* on embryonic precursor mesenchymal cells of the sea urchin was further confirmed by *in situ* analyses (Meyer et al., 2023 [↗](#)). In addition, this latter group also reported these mesenchymal cells to express a GATA transcription factor (*GATA3*) and *ETS1*, which in our dataset are also being expressed only by populations within this supra-cluster (*GATA2* and *ETS1* – **Fig 3B** [↗](#)). The gene *ETS1* has been associated with the differentiating sea urchin larvae mesenchymal cells and has a vital role in EMT and cellular invasion of mesenchymal cancer cells (Gluck et al., 2019 [↗](#); Koga et al., 2010 [↗](#); Meyer et al., 2023 [↗](#)). The expression of *PRG4* (proteoglycan-4) reinforces the mesenchymal identity of this supra-cluster in the regenerating intestine (**S4 Fig** [↗](#)), which correlates with previous studies that have identified a proteoglycan-like molecule in the mesenchyme of 7-dpe regenerating intestine (Vázquez-Vélez et al., 2016 [↗](#)).

In contrast, the top marker gene in the other supra-cluster is *AHNK*, known as neuroblast differentiation-associated protein AHNK. Reports have shown *AHNK* to have a role in calcium regulation, cellular migration, and carcinogenic transformation of colon epithelial cells (Dumitru et al., 2013 [↗](#)). Furthermore, this gene is overexpressed in regenerating rat muscle compared to normal muscle (Huang et al., 2007 [↗](#)). However, the localized expression of *WNT9* in the C1, C3, C5, and C8 of this supra-cluster more clearly favors its classification as a marker for coelomic epithelial cell types (**Fig 3B** [↗](#) and **S4 Fig** [↗](#)). A previous study from our laboratory using *in situ* hybridization, details the expression of *WNT9* during intestinal regeneration in *H. glaberrima*, where it was shown to be localized to the coelomic epithelium of the anlage and adjacent mesentery (Mashanov et al., 2012 [↗](#)). In addition, correlating with what was shown by the *in situ* hybridization results, the population of cells that differentially expresses *WNT9* makes 15% of those in the mesentery. However, it is close to 30% of the cells in the anlage. Additional analyses, discussed below, further strengthens the coelomic epithelium identity of cells in this supra-cluster.

In summary, results show 13 individual cell clusters with distinct expression profiles in the regenerative intestinal tissue. Moreover, our results suggest a principal separation of clusters based on whether they correspond to those of the mesenchyme or to cells from the coelomic epithelia layer. The rest of the populations show top expressed genes that are immune-related suggesting that these must be coelomocyte populations (**Fig 2** [↗](#)).

Cluster Identities

Rather than considering single genes, the uniqueness of each cluster can be assessed in terms of their transcriptomic profile (**Fig 3A** [↗](#)) and by the expression of transcription factors and intercellular signaling molecules (**Fig 3B** [↗](#)). As shown in **Fig 3A-B** [↗](#), each cluster has a unique transcriptomic profile. Nonetheless, some clusters show their close relationship (epithelial vs mesenchymal) based on shared gene expression or representation. A notable example is clusters C0, C1, C3, and C4 of the epithelial layer populations, which share many top genes, albeit at different expression levels. Similarly, C8 and C9 have low expression of genes expressed by C0, C1, C3 and C4 but have expression of other genes that are not expressed by any other cluster. This is also true for the mesenchymal clusters, where these two clusters have overlap of some genes, that are not expressed by other clusters. Therefore, these results show the interaction of these clusters and their transcriptomic relationship depending on where they are localized within the regenerating intestinal tissue.

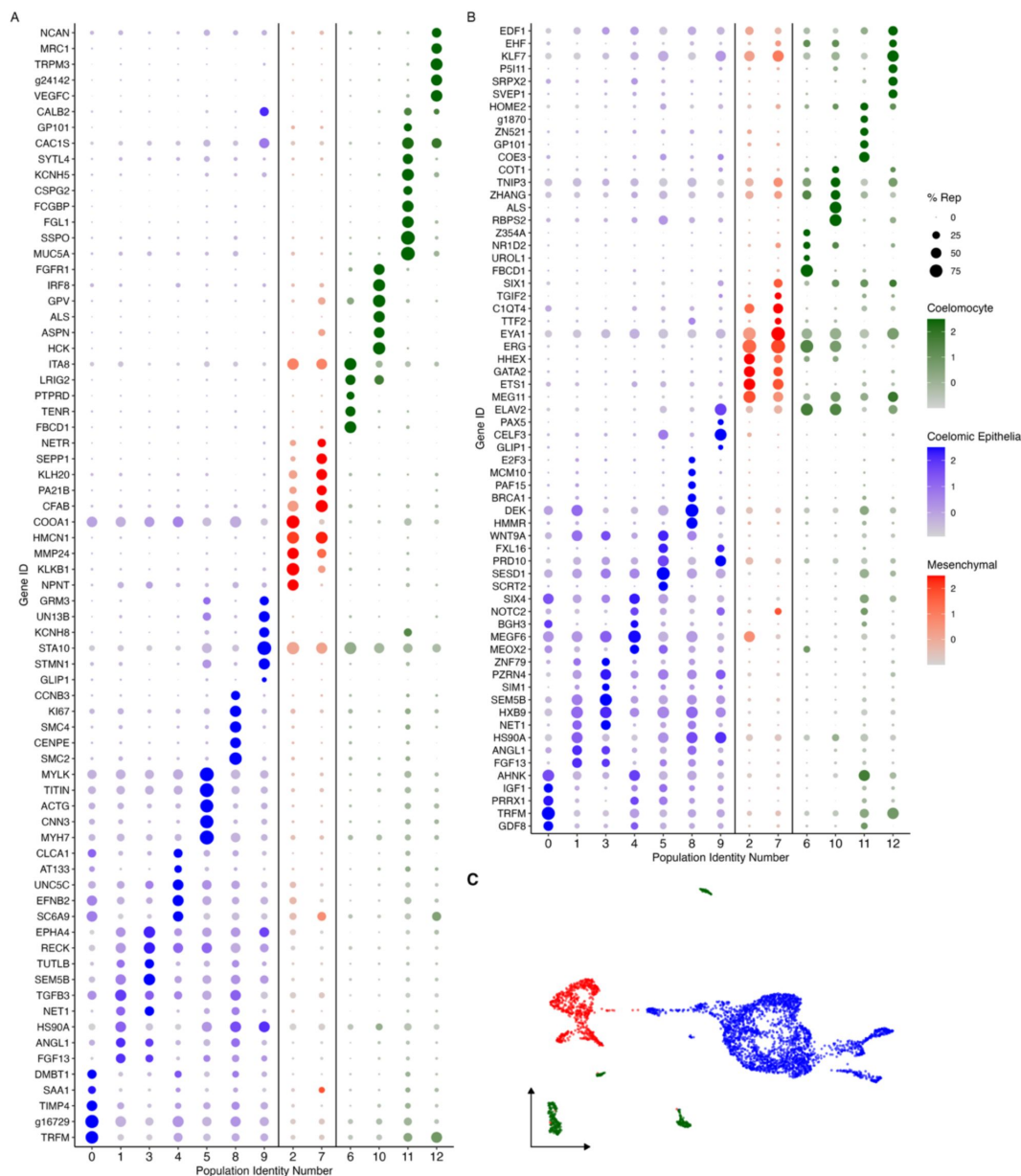


Fig 3.

Cluster characterization by gene expression.

(A) Top-expressed genes or (B) genes corresponding to transcription factors and intercellular signaling molecules are identified in the 13 cell clusters. Clusters are classified by their corresponding cell type, where blue corresponds to cells of the coelomic epithelium, red to those in the mesenchyme and green to coelomocytes. Color intensity shows the expression level of each gene in log2fold-change (log2FC) values. Dot size corresponds to percentage of representation of the gene in the respective cluster compared to all others. Gene identifiers starting with "g" correspond to uncharacterized gene models of *H. glaberrima*. (C) UMAP plot of clusters colored by cell type.

Identifying the top genes expressed by each cluster also provides essential, notwithstanding limited, information for their complete identification. To further characterize each cluster, we have used other analyses or performed additional experiments, including (1) multiple gene expression patterns, (2) enriched ontology, (3) HCR-FISH and (4) pseudo-trajectory. Initially, we describe in depth the potential identity of these clusters based on their expression pattern and enriched ontology, and we will end with their potential interactions.

An in-depth analysis of the various clusters and their possible relation to cell populations

Coelomocyte populations

Coelomocytes are specialized cells found in the coelomic fluid and within organs of echinoderms. These cells have been associated with immunological roles including pathogen recognition, encapsulation, phagocytosis, debris removal, cytokine production and secretion among others (Barela Hudgell et al., 2022; Courtney Smith et al., 2018; [García-Arrarás et al., 2006](#); [Smith et al., 1995](#)). In holothurians, these cells have been shown to be present at injury sites and in the regeneration anlage. Coelomocytes can be subdivided into different populations by using morphological, physiological, and molecular characteristics ([Ramírez-Gómez et al., 2010](#)).

Many of the coelomocyte characteristics correlate with the top differentially expressed genes of cell populations within our data. These immune-like clusters (C6, C10, C11, and C12) have high expression of genes related to the immune system that are not shared with any other cluster. For example, C6 embodies a distinct cell population that represents a substantial number of cells (6% of all cells, 8% of the mesentery and 4% of the cells in the anlage). These cells are the only population expressing *FBCD1* (fibrinogen C domain-containing protein 1). Cells in C6 also express other genes such as various tyrosine protein phosphate receptors, integrin alpha-8 (*ITA8*), platelet glycoprotein V (*GPV*), leucine rich repeats and immunoglobulin-like domains protein 2 (*LRIG2*) ([Fig 4A](#)). The immune identity of C6 is also supported by the resulting gene set enrichment of gene ontology (gseGO) terms related to immune responses, such as *ubiquitin-dependent ERAD pathway*, *innate immune response activating cell surface receptor signaling pathway*, *respond to endoplasmic reticulum stress*, *phagocytosis* and many more related to defense mechanisms ([Fig 4B](#)). Similarly, C10 and C12 transcriptomic profiles suggest they correspond to immune-like populations ([Fig 4C](#)). Specifically, C10 shows various uncharacterized genes along with *HCK* (tyrosine-protein kinase HCK), *DMBT1* (deleted in malignant brain tumors 1), *ALS* (insulin-like growth factor binding protein complex acid labile subunit), *GPV*, and *IRF8* (interferon regulatory factor 8), *FER* (tyrosine-protein kinase Fer). The top GO enriched terms of this cluster support its involvement in immune process, some of these being: *lipase activity regulation*, *Fc receptor mediated stimulatory signaling pathway*, *cellular pigmentation*, *B cell activation involved in immune response*, and *Fc receptor signaling pathway*. The transcripts expressed by C12 show a more complex profile, where most of the top genes are uncharacterized. Nonetheless, among the annotated genes are *BAR3* (Balbiani ring protein 3), *LYS1* (lysozyme 1), *TRPM3* (transient receptor cation channel subfamily M member 3), *PGCA* (aggrecan core protein) and *MRC1* (macrophage mannose receptor 1). The top genes of C11 include a great number of immune genes such as *MUC5A* (mucin-5AC), *FCGBP* (IgG-Fc-binding protein), *SSPO* (SCO-spondin), *FCN1A* (ficolin-1-A), *MUC5B* (mucin-5B), and *TIE1* (tyrosine-protein kinase receptor Tie-1). However, the top genes of this cluster also include several genes involved in neuronal activity, which is evident in the top enriched GO terms of this cluster that include *regulation of postsynaptic membrane potential*, *excitatory postsynaptic potential*, *chemical synaptic transmission*, *endoplasmic reticulum to Golgi vesicle-mediated transport*, *adult behavior*, and *regulation of neurotransmitter levels* ([Fig 4A-B](#)).

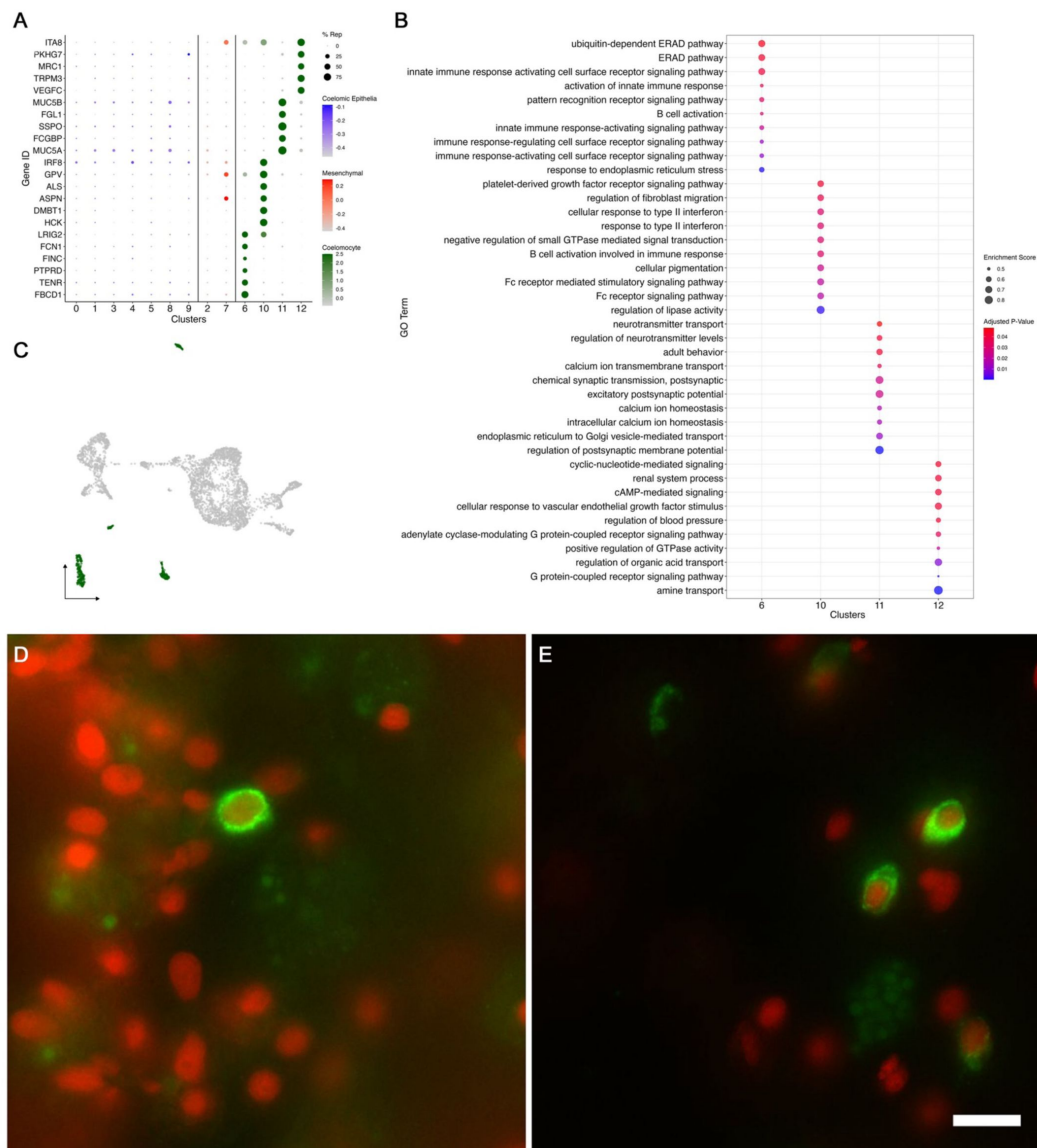


Fig 4.

Expression profile of coelomocyte cell types.

(A) Dot plot showing the top expressed genes of C6, C10, C11, and C12. Color intensity reflects the expression levels in log2FC values. Dot size correspond to the percentage of representation of each gene in the respective cluster compared to all others. (B) Top enriched GO biological processes terms of C6, C10, C11, and C12. Dot size corresponds to the enrichment score while the color reflects the adjusted p-value. (C) UMAP highlighting the coelomocyte populations. (D-E) HCR-FISH for *FBCD1* in holothurian intestinal tissues. Cells expressing *FBCD1* mRNA in the connective tissue layer of (D) regenerating mesentery and (E) normal intestine of *H. glaberrima* are shown. Red = DAPI stain, green = HCR-FISH. Bar= 10 μ m.

To partially confirm the prediction that cells from C6, C10, C11 and C12 corresponded to coelomocyte populations, we used HCR-FISH to identify the cell types expressing the top gene in one of the clusters. We focused on the expression of *FBCD1*, the gene that is the most represented by cells of C6. *In situ* hybridization identified a distinct cell type in both regenerating and non-regenerating tissues (**Fig 4D-E**). These cells showed round or oval morphologies with a central round nucleus. In some cases, short extensions could be observed. The cells were heterogeneously distributed in all tissues, including the nervous system and the body wall, and could be found associated with either epithelial tissues or with the extracellular matrix (ECM) in the normal intestine, the intestinal anlage and in the mesentery of normal or regenerating animals. The widespread distribution of this cell type hinted at a cell function consistent with patrolling the body to detect and respond to potential threats such as injury or bacterial invasion.

Mesenchymal cell populations

Mesenchymal cells of the intestinal anlage are yet to be well studied. They are known to be less proliferative than those in the overlying epithelium (García-Arrarás et al., 2011; 1998) and involved in ECM remodeling (Quiñones et al., 2002). Some of the mesenchymal cells are thought to migrate from the connective tissue of the mesentery (Cabrera-Serrano and García-Arrarás, 2004), while others have been shown to originate via EMT from the overlying epithelium (García-Arrarás et al., 2011). Some of these cells will form a mesenchymal cellular layer associated with the luminal epithelial cells as the lumen forms (García-Arrarás et al., 2011). As explained earlier, we have proposed that C2 and C7, form a separate supra-cluster, clearly identified by the increased expression of *ERG* with 5 of log₂FC and a 90% representation (**Fig 20**). These two clusters also share many marker genes, including multiple ECM genes (**Fig 5D**). For instance, highest expressed gene of C2 is *TIMP3* (metalloproteinase inhibitor 3), followed by *NPNT* (nephronectin), which has been reported to be an integrin ligand during kidney development (Sun et al., 2018). Additionally, this cluster has overexpression of *ECM2* (extracellular matrix protein 2), *KLKB1* (plasma kallikrein), *MMP14* (matrix metalloproteinase 14), *MMP24* (matrix metalloproteinase 24), *HMCN1* (hemicentin-1), and *ITA8* (integrin alpha-8), all of which are explicitly related to extracellular matrix component (Bokel and Brown, 2002; Dolmatov and Nizhnichenko, 2023; Dong et al., 2006; Stamenkovic, 2003; Volkert et al., 2014). Many of these genes are also highly represented in C7. However, here we also find *ITIH2* (inter-alpha-trypsin inhibitor heavy chain H2), *DUOX1* (dual oxidase 1), *SVEP1* (sushi, von Willebrand factor type A, EGF and pentraxin domain-containing protein 1), *SEPP1* (selenoprotein P), and *KLH20* (Kelch-like protein 20) suggesting that cells have gained some specialization and are more advanced in their differentiation when compared to those of C2. Along with this, when analyzing the gseGO, for C2 we obtain GO terms of numerous ECM processes, such as *cell adhesion mediated by integrin*, *integrin-mediated signaling pathway*, and *regulation of cell-substrate junction assembly* (**Fig 5F**). Similarly, C7 has an enrichment of *regulation of cell-matrix adhesion*, *regulation of cell-substrate junction organization*, and *substrate adhesion-dependent cell spreading*. Notably, these clusters also show enrichment of processes involved in cell proliferation, growth, and wound healing. To mention a few for C2 we find *B cell differentiation*, *vascular endothelial growth factor receptor signaling*, *regulation of myeloid leukocyte differentiation*, and *negative regulation of myeloid cell differentiation*, while C7 shows an enrichment of *positive regulation of wound healing*, *regulation of neural precursor cell proliferation*, *positive regulation of response to wounding*, *humoral immune response*, *neuroblast proliferation*, among others. This is not surprising considering that this tissue is undergoing active regeneration which requires growth and healing events to be active at distinct stages of the process.

To confirm our prediction that cells in C2 and C7 were those present within the mesenchyme, we chose to localize the expression of *HMCNT1* mRNA. The mRNA for this protein, known to code for an extracellular protein (Dolmatov and Nizhnichenko, 2023; Lindsay-Mosher et al., 2020; Welcker et al., 2021), is present in both cell clusters. HCR-FISH showed that cells expressing the *HMCNT1* mRNA were present in the connective tissue of the regenerating mesentery and certain

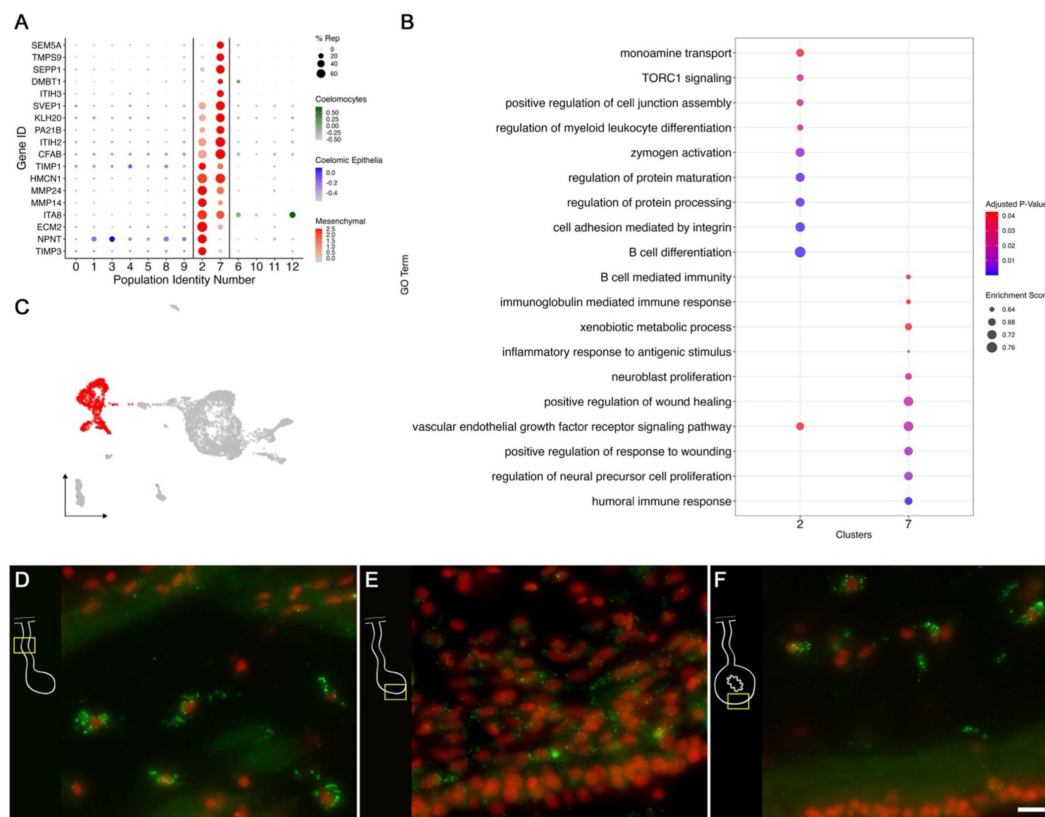


Fig 5.

Expression profile of mesenchymal cell types.

(A) Dot plot showing the top expressed genes of C2 and C7. Color intensity reflects expression levels in log2FC values. Dot size correspond to the percentage of representation of each gene in the respective cluster compared to all others. (B) Top GO enriched terms of biological processes of C2 and C7. Dot size corresponds to the enrichment score while the color reflects the adjusted p-value. (C) UMAP highlighting the clusters that correspond to the mesenchymal population (D-F) Cells expressing *HMCNT1* mRNA in the connective tissue layer of (D) regenerating mesentery, (E) intestinal anlage and (F) normal intestine of *H. glaberrima*. Red = DAPI stain, green = HCR-FISH. Bar= 10µm

regions of the connective tissue of the anlage. In the normal intestine, labeled cells were present in the connective tissue (**Fig 5A-C**). Cells in the mesentery and normal intestine connective tissue were somewhat distanced from each other and had an irregular morphology with intense labeling throughout the cytoplasm. A weaker labeling was observed in cells of the anlage, which were more densely packed and adjacent to the coelomic epithelium, suggesting that they correspond to cells undergoing EMT on their way to differentiate into ECM-producing mesenchymal cells. A very similar pattern of expression was observed with HCR-FISH for *ERG* (not shown). The *ERG*-expressing cells were also found in the connective tissue of normal intestine and in the cells of the anlage undergoing EMT.

Coelomic epithelia/mesothelial cell populations

The mesothelial layer of the intestine and mesentery is formed by cells in contact with the coelomic fluid (coelomic epithelium or peritoneocytes) together with myocytes and neurons. In the regenerating tissues, some of these cells dedifferentiate and form a coelomic epithelia that differs in morphology (Mashanov et al., 2005) and gene expression (Mashanov et al., 2017, 2015, 2012, 2010) to the mesothelium that normally surrounds the organ. This dedifferentiated coelomic epithelium which is present mainly in the anlage and in areas of the adjacent mesentery, is also responsible for most of the cell division that takes place in the regenerating intestine (García-Arrarás et al., 2011). Many of the changes observed in the mesothelium during regeneration take place in a gradient, beginning at the tip of the mesentery (where the anlage forms) and continuing in the adjacent regions of the mesentery mesothelium. Thus, the analysis of the scRNA-seq data in view of our knowledge of the ongoing events in the 9-dpe regenerating organ leads to the conclusion, as stated previously, that the 7 clusters within the major supra-cluster, represent the cells in the coelomic epithelium of the mesentery and the anlage. Here is our analysis:

C8 represents the proliferating cells

These cells mainly found within the anlage coelomic epithelia express proliferation markers such as *PLK1* (serine/threonine-protein kinase PLK1), *SMC2* (structural maintenance of chromosomes protein 2), *PRI2* (PRIM2 - DNA primase large subunit), *PCNA* (proliferating cell nuclear antigen), and *CDK1* (cyclin-dependent kinase 1) (Locard-Paulet et al., 2022). In addition, this cluster has high expression of *TOP2A* (DNA topoisomerase 2-beta), a gene that has also been seen to be overexpressed in proliferating basal cells of the human gastrointestinal epithelia (Busslinger et al., 2021). Other marker genes related to cell mitotic activity found here include *CENPE* (centromere-associated protein E), *SMC4* (structural maintenance of chromosomes protein 4), *K167* (proliferation marker protein Ki-67), and *CCNB3* (G2/mitotic-specific cyclin-B3) (**Fig 6A**). Furthermore, this cluster expresses specific transcription factors such as *E2F3*, *MCM10* (protein MCM10 homolog), *PAF15* (PCNA-associated factor) and *BRCA1* (breast cancer type 1 susceptibility protein) that are also associated with control of cell division (Humbert et al., 2000; Lööke et al., 2017; Xie et al., 2014). The gseGO terms also confirm its proliferative identity with GO terms related to *chromosome separation*, *condensation*, *mitotic cytokinesis*, and *regulation of cell cycle checkpoint* (**Fig 6B**). Moreover, the dividing cell population is higher in the anlage samples (4%) than in the mesentery samples (2%), which is in accordance with what we have observed in regenerating animals, that while cell division does take place in the mesentery, more cells are proliferating in the epithelial layer of the intestinal anlage (Bello et al., 2020; García-Arrarás et al., 2011, 1998).

C5 represents muscle precursors

A second cell population known to originate from the coelomic epithelium and that can be identified in our data is that of muscle precursors. These cells are known to differentiate into enteric muscle during the second week of regeneration (Murray and García-Arrarás, 2004). This population can be recognized in our data by the expression of muscle-specific markers present in

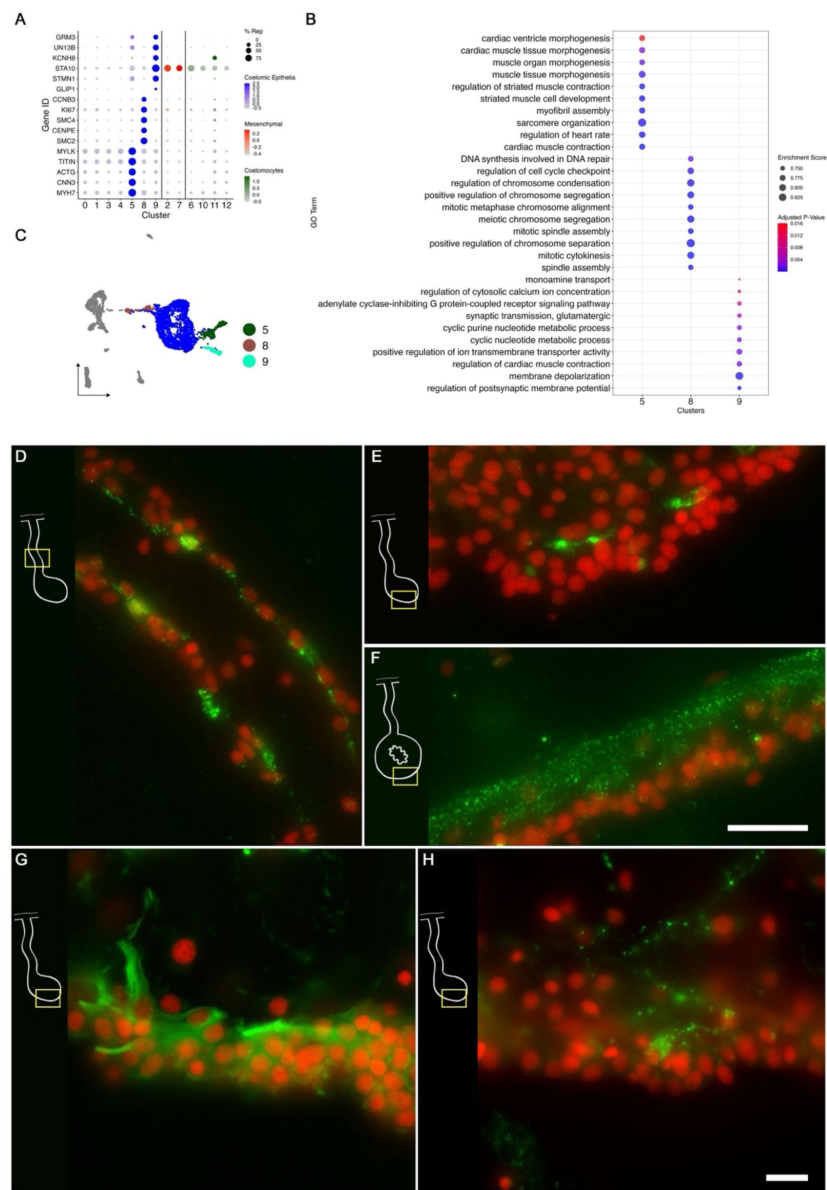


Fig 6.

Expression profile of cell clusters undergoing differentiation or proliferation in the coelomic epithelium.

(A) Dot plot showing the expression profile of C5, C8, and C9. Color intensity reflects expression levels in $\log_2FC$ values. Dot size represents percentage of representation of each gene. (B) Top GO enriched terms of biological processes of each cluster. (C) UMAP highlighting C5, C8, and C9, which along with clusters in color blue correspond to the coelomic epithelium. (D-F) Cells expressing *MYH7* mRNA in the (D) regenerating mesentery, (E) basal area of the anlage coelomic epithelium and (F) muscle layer of the normal intestine. (G) Phalloidin labeled differentiating muscle cells in basal area of the coelomic epithelium. (H) Cell labeled with RN1 antibody (neuronal marker) in coelomic epithelia of the anlage. Red = DAPI stain, green = HCR-FISH. G= Fluorescent Phalloidin, H= RN1 antibody. Bar-D-F= 25 μ m, G-H=10 μ m.

C5, such as *TITIN*, *MYL1* (myosin light chain 1/3), *MYH7* (myosin 7), *ACTG* (actin, cytoplasmic 2), and *CNN3* (calponin-3) (**Fig 6A**). Considering that this cell population is potentially a population of the coelomic epithelium actively undergoing differentiation toward muscle phenotype, they also share gene expression with other epithelial cell clusters (C0, C1, C3, C4), albeit at a lower fold-change. Similarly, this cluster shows a specific expression of transcription factors associated to muscle cells, namely *FXL16* (F-box/LRR-repeat protein 16) and *SCRT2* (transcriptional repressor scratch 2). Results of GO terms of this cluster also demonstrate enriched terms related to muscle tissue growth, such as *muscle tissue morphogenesis*, *muscle development*, *myofibril assembly*, and *sarcomere organization* (**Fig 6B**). The focus on development and morphogenesis is to be expected, considering that these are still undergoing differentiation toward a muscle phenotype. HCR-FISH of *MYH7* corroborates the muscle precursor phenotype of the cells in this cluster (**Fig 6D-F**). Labeling is observed in the same regions as myoblasts or muscle cells were previously identified (Murray and García-Arrarás, 2004), including differentiating muscle cells within or underlying the coelomic epithelia of the mesentery and the anlage (**Fig 6D & 6E**) and, as expected, the muscle layer in the normal intestine (**Fig 6F**). The same cell population found in the basal side of the coelomic epithelia of the regenerating intestine is labeled with the muscle marker, fluorescently-labeled phalloidin as shown in **Fig 6G**.

C9 represents the neuroepithelial cells

Cells in C9, represent another population of specialized cells that can be associated to cells previously described in the intestinal anlage. This small number (3%) of cells most likely corresponds to neuroepithelial cells that will eventually give rise to neurons. These cells are shown to be expressing neuroepithelial or neuronal genes such as neurotrophin (*NETR*), potassium gated-voltage channels, *PRD10* (*PRDM10* - PR domain zinc finger protein 10), *ELAV2* and *STA10* (*STARD10* - START domain-containing protein 10). The latter is a protein that we have characterized as being expressed by enteric neurons and nerve bundles (Rosado-Olivieri et al., 2017) (**Fig 6A**). The holothurian STA10 (*STARD10*) is recognized by our monoclonal antibody RN1. This antibody has been used to detect enteric neurons as they begin to differentiate in the coelomic epithelium during the second week of regeneration (Tossas et al., 2014). Labeling of the 8-dpe regenerating mesentery shows one of these RN1 neuronal cells in the coelomic epithelium of the regenerating anlage (**Fig 6H**). Among its associated GO terms are *positive regulation of ion transmembrane transporter activity*, *cyclic nucleotide metabolic process*, *regulation of muscle contraction*, *regulation of membrane potential* and *positive regulation of hormone secretion* (**Fig 6B**). Moreover, there is also a great representation of processes involved in development, differentiation, and growth of nerve cells, all of which together would be expected of a neuroepithelial layer.

The characterization of these 3 clusters, that represent cells undergoing differentiation or proliferation, leaves a group of 4 clusters (C0, C1, C3, and C4) that show some overlap in expressed genes and at the same time share some gene expression with some of the previously described clusters. Nonetheless, as seen in **Fig 3A & B**, the cells in these four clusters still have high expression and representation of specific transcripts.

C4 represents the intestinal coelomic epithelial cells

The gene expression profile of cells in C4 sets them slightly apart from the other three clusters (C0, C1, C3). It identifies cells that are more advanced in their development toward a particular phenotype. C4 shows high expression of genes such *KCNQ5* (potassium voltage-gated channel subfamily KQT member 5), *SC6A9* (sodium and chloride dependent glycine transporter 1), *EFNB2* (Ephrin-B2), and *UNC5C* (Netrin receptor UNC5C) (**Fig 7A**). Other than these, C4 also shows high expression of genes that are related to cell-cell interactions and ECM molecules such as *LAMA2* (laminin subunit alpha-2), *MEGF6* (multiple epidermal growth factor-like domains protein 6), *FMN1* (formin-1), *NPHN* (nephrin). This cluster, distinct from others, shows enrichment of GO terms related to more advanced stages of development, such as *morphogenesis of epithelium*,

sensory perception, regulation of calcium ion transmembrane transport, among others (**Fig 7B** [↗](#)). Noteworthy, cells from this cluster correspond mostly to cell from the mesentery samples (67%) rather than from the anlage (37%). Additionally, from all the mesentery cells, about 11% are part of this cluster, while only 4% of the anlage cells are represented here. HCR-FISH of *SC6A5* (sodium- and chloride-dependent glycine transporter 2), a gene differentially expressed in the cells of this cluster provided a surprising result. The expression of this mRNA was observed in a few cells in the coelomic epithelia of the regenerating mesentery and intestinal anlage (**Fig 7D-E** [↗](#)). However, intense labeling is observed in some cells of the normal intestine mesothelium. This pattern of labeling strongly suggests that the labeled cells of the regenerating intestine grouped in C4 (both mesenteral and anlage) correspond to cells that are in a differentiation pathway to become part of the coelomic epithelia (possibly the peritoneocytes) in the regenerated organ. This conclusion is strengthened by the pseudotime analyses presented in the following section.

C0, C1, and C3 represent differentiation stages of coelomic epithelial cells

The three remaining clusters to be analyzed are C0, C1, and C3. These three clusters share many of their top representative genes, which are associated with developmental, regenerative, or oncogenic processes (Bradford et al., 2009 [↗](#); Dunn et al., 2006 [↗](#); Lu et al., 2015 [↗](#); Nishimoto and Nishida, 2007 [↗](#); Oike et al., 2004 [↗](#); Zhao et al., 2008 [↗](#)). These include, for cluster 0: *TRFM* (melanotransferrin), *TIMP4* (metalloproteinase inhibitor 4), and *DMBT1*, for cluster 1: *FGF13* (fibroblast growth factor 13), *TGFB3* (transforming growth factor beta 3), *HS90A* (heat shock protein HSP 90-alpha), and *ANG1* (angiopoietin-related protein 1) (**Fig 7A** [↗](#)), and for cluster 3: *SEM5B* (semaphoring-5B), *LRIG3* (leucine-rich repeats and immunoglobulin-like domains protein 3), *TUTLB* (protein turtle homolog B), and *NET1* (netrin-1). Moreover, C0 and C1 share an over-representation of ribosomal genes (not shown), and this is reflected in their GO analyses that highlights biological processes related to ribosomal activity such as *cytoplasmic translation*, and *ribosome assembly, biogenesis, and assembly* (**Fig 7B** [↗](#)). In addition to these, C3 also showed enrichment of processes involved in the development of tubular lumen-containing structures such as *mesonephric and ureteric ducts, differentiation and regulation of cell growth*, and *negative regulation of axogenesis* (**Fig 7B** [↗](#)).

C1 and C3 are also closely related in their localization, being overwhelmingly associated with the anlage. These two clusters are mostly composed of cells from the anlage tissue, where we expect to see the precursor cells that will give rise to specialized cells of the organ (**Fig 1D** [↗](#)). Precisely, C1 and C3 cells together correspond to 37% of all anlage cells, compared to 14.6% of all mesentery cells. In contrast, cells from C0 correspond to 27% and 18% of the mesentery and anlage cells, respectively.

To obtain insight into these cell clusters, we performed HCR-FISH for two different mRNAs; *NET1*, a chemotropic protein highly represented in C1 and C3, and *TRFM*, the top represented gene in C0. Both *in situ* hybridization experiments labeled cells in the coelomic epithelium of the regenerating intestine, supporting our contention that the large supra-cluster represents the coelomic epithelium layer. However, their spatial pattern of expression was unpredictably different. While *NET1* was highly expressed in most of the coelomic epithelial cells of the anlage, little expression was found in the regenerating mesentery or in the coelomic epithelium of the normal intestine (**Fig 8G-H** [↗](#)). *TRFM*, in contrast, was highly expressed in the coelomic epithelium of the normal intestine and poorly expressed in the intestinal anlage (**Fig 8A-E** [↗](#)). In the regenerating mesentery, a gradient in expression of the *TRFM* is observed, where high levels of expression were found in the coelomic epithelium close to the body wall and diminished as one approached the anlage. Thus, the HCR-FISH results show that C0, C1, and C3 correspond to cells of the coelomic

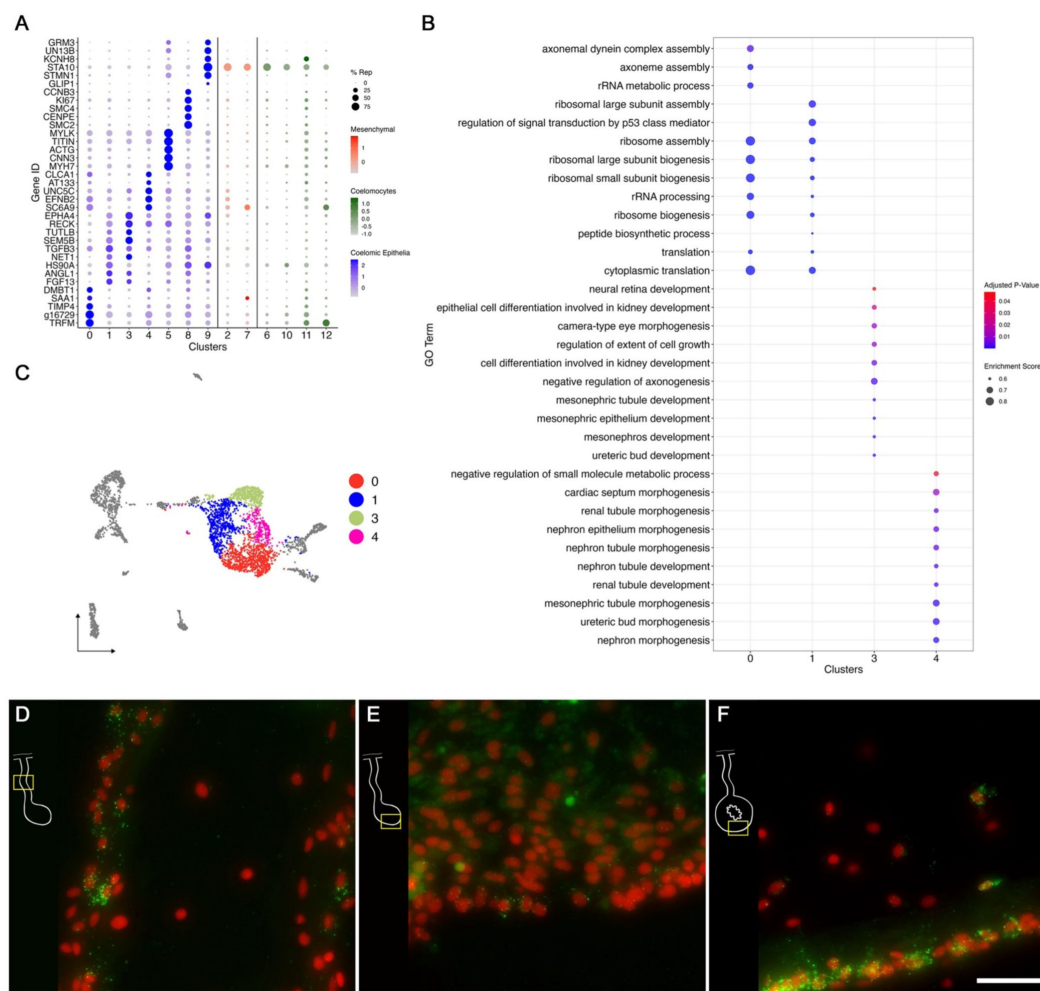


Fig 7.

Expression profile of coelomic cell populations in the regenerating intestine anlage and mesentery.

(A) Dot plot showing the top expressed genes of C0, C1, C3 and C4. Color intensity reflects expression levels in log2FC values. Dot size correspond to the percentage of representation of each gene in the respective cluster compared to all others. (B) Top GO enriched terms of biological processes of each cluster. Dot size corresponds to the enrichment score while the color reflects the adjusted p-value. (C) UMAP highlighting the clusters of the coelomic epithelial cells of the regenerating intestine. (D-E) HCR-FISH for *SC645* mRNA was performed as a marker for cells of C4. Labeling is observed in the (D) regenerating mesentery, (E) the coelomic epithelium of the anlage and (F) coelomic epithelium of the normal intestine. Red = DAPI stain, green = HCR-FISH. Bar= 25µm.

epithelium, but strongly suggest that C0 differs from C1 and C3 both in their gene expression profile and in the localization where they are found, both in the regenerating and in the normal intestine.

Finally, it is essential to highlight the many signaling, or growth factors expressed by the coelomic epithelial clusters, in particular C0, C1 and C3 (**Fig 3**). These include: Wnt, Hox, semaphorin, FGFs, TGF-beta, netrin, insulin-like growth factor (IGF), growth/differentiation factors (GDF), and angiopoietin related proteins (ANG1) (e.g. *WNT9*, *SEM5B*, *FGF13*, *NKx3.2*, *TGFB3*, *HOX9*, *IGF1*, *GDF8*, *ANG1*, and *FOXF1*) (**Fig 3**). This is important in view that the epithelium of the vertebrate blastema is characterized by its chemical modulation of the underlying mesenchyme, as will be discussed later. Likewise, other genes that serve as markers of specific cell types or cellular stages were also identified, including *PIW1* (piwi-like protein 1) in C1, *YAP1* in C3 and C4, *HES1* (transcription factor HES-1) in C1 and C3, and *PRRX1* (paired mesoderm homeobox protein) in C0 and C4 (**Fig 3** and **S5 Fig**).

In summary, of the 13 clusters identified, our data strongly suggest that four of them (C9, C10, C11, and C12) correspond to coelomocytes or immune cells, two of them (C2 and C7) correspond to cells with a mesenchymal phenotype and the remaining 7 to cells of the coelomic epithelia. Of these seven, C5 corresponds to differentiating muscle, C9 to differentiating neuroepithelium, C4 to differentiating coelomic epithelia and C8 to proliferating cells. C1 and C3 represent most of the cells found in the coelomic epithelium of the anlage. In contrast, C0 represents a coelomic epithelial phenotype more closely associated with the mesentery than the intestine.

Trajectory Analysis: What populations are driving cell specification?

Among the many mysteries of the holothurian intestinal regeneration process is the identification of the precursor cells. In simple terms, what are the cells from which all nascent cells derive from? We performed a trajectory analysis of the data to provide some insights into this issue. This type of analysis is usually performed with samples at different stages or time points. However, we considered it feasible to conduct this analysis because in the 9-dpe regenerating anlage/mesentery we find cells at various stages of differentiation. These cells could provide crucial information on how the cell populations are associated with each other. To address this, we initially employed RNA velocity analysis. This method describes the temporal dynamics of gene expression based on the relative abundances of spliced and unspliced mRNA across cell populations.

Our initial velocity analysis on all the clusters and samples (**Fig 9A-B**), provided three main results. First, the direction of arrows in our UMAP shows them flowing towards C5, C9, and C4. These arrows do not point towards any other cluster; thus, they are terminal arrows. Second, while arrows from C8 are not terminal, they are directed towards C1. This suggests that cells in C1 provide cells for the growth of the anlage via proliferation. Thus, these results support C5, C9 and C4 as terminal cell clusters which we have described as muscle, neuroepithelial, and the nascent coelomic epithelium cells, respectively. Third, velocity embedding shows shorter arrows that point from clusters 0 and 1 towards terminal populations previously described. Therefore, cells of C0 and C1 are not undergoing significant transcriptional changes. The RNA velocity results of the rest of the clusters are less interesting as they do not show directions towards any other clusters, mainly because of their individuality within the UMAP. However, it is interesting that arrows of the mesenchymal cell populations show distinct directions and lengths. Based on the results, it seems that portions of both mesenchymal clusters (C2 and C7) have gone or are undergoing more extensive differentiation changes.

We then re-clustered C0, C1, and C3 cells, along with the differentiating cell populations (C4, C5, and C9) to better understand their relationship (**Fig 9C**). The velocity assessment of these newly clustered populations resulted in a similar pattern (**Fig 9D**). Here, we can see that the

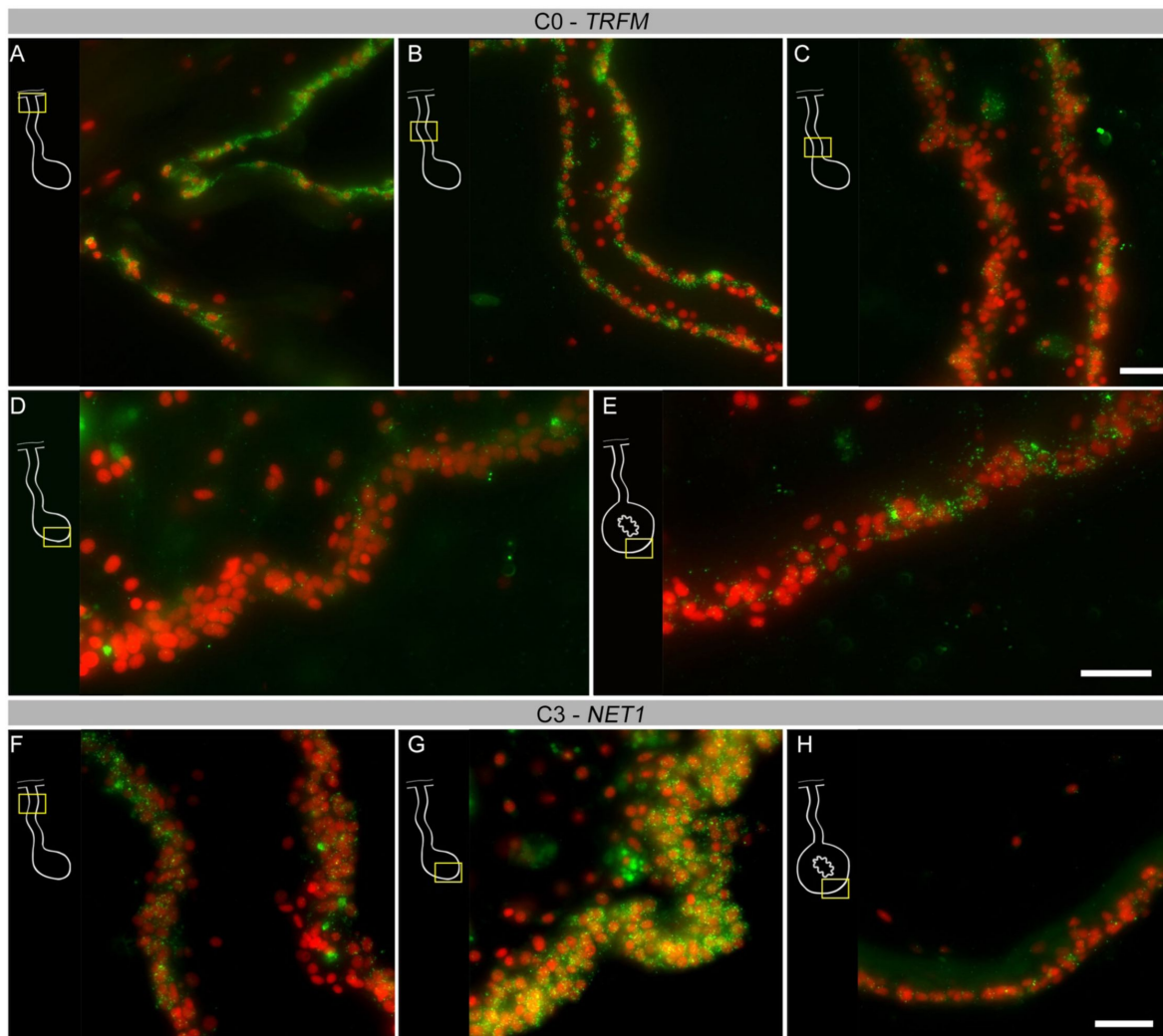


Fig 8.

HCR-FISH for *TRFM* and *NET1* shows the differential gene expression of the anlage versus mesentery coelomic epithelia.

TRFM is mainly expressed in the mesentery coelomic epithelial cells that are distant from the regenerating anlage while *NET1* is mainly expressed in the coelomic epithelial cells of the anlage. *TRFM* mRNA expression (A-E) shows a gradient from (A) very strong expression in the coelomic epithelium at the attachment of the mesentery to the body wall, (B) high expression in the coelomic epithelia of the mesentery close to the body wall, (C) weaker expression in the mesentery close to the anlage to (D) weak or lack of expression in the coelomic epithelium of the anlage. (E) *TRFM* mRNA is expressed by the coelomic epithelium of the normal intestine. regenerating intestine. In contrast, few cells express (A) *NET1* mRNA in the regenerating mesentery, while high expression is found in (B) the coelomic epithelium of the anlage (C) Little expression of *NET1* is present in the coelomic epithelium of the normal intestine. Red = DAPI stain, green = HCR-FISH. Bar= 25µm.

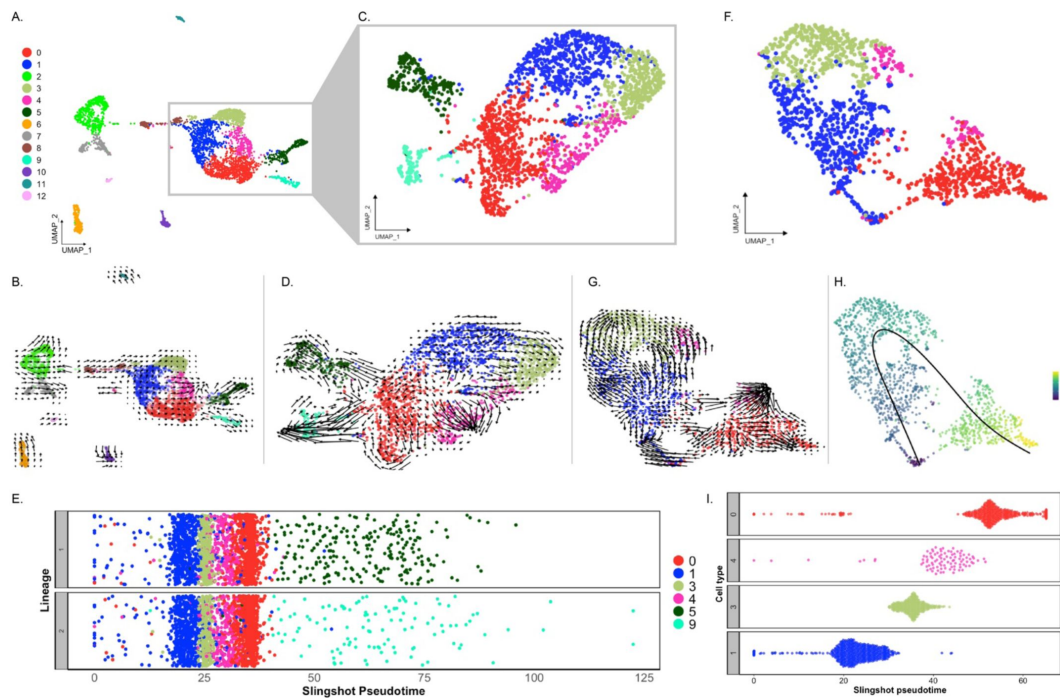


Fig 9.

Trajectory analysis of cell populations from the regenerating intestinal tissue.

(A) UMAP plot of all identified clusters. (B) RNA velocity embedded in UMAP of all main clusters. (C) UMAP of re-clustering of cells from C0, C1, C3, C4, C5, and C9. (D) RNA velocity analysis results from the subset from panel (C). (E) Jitter plot of Slingshot pseudotime of cells from C0, C1, C3, C4, C5 and C9. Pseudotime resulted in two lineages one containing C5 and the other C9. (F) UMAP of re-clustered anlage cells corresponding to C0, C1, C3, and C4. (G) RNA velocity results from the cluster results from panel F. (H) UMAP of panel F plot overlaid with pseudotime results of Slingshot. Color represent pseudotime values from 0 (blue) to 60 (yellow). (I) Jitter plot showing the Slingshot pseudotime of cells from each cell cluster.

differentiating cells (C4, C5, and C9) have long arrows suggesting that these cells are going through an advanced stage of transcriptional change compared to others to some extent, C4. Moreover, it seems that C0 has a closer relationship to cells of C5 and C9 and that some cells of these clusters could potentially differentiate into cells of C4 (the coelomic epithelium cells). In this case, to complement and confirm our RNA velocity interpretation, we also performed a pseudotime analysis using Slingshot, which relies on the expression data of each cluster. This analysis showed C1 in an earlier pseudotime than C3, C4 and C0 in the resulting two lineages (**Fig 9D** [↗](#)). The resulting lineages differed by the terminal clusters, one containing C5 (muscle) and the other C9 (neuroepithelial). Thus, it supports what we have already visualized on the RNA velocity embeddings.

The results described so far show that C0, C1 and C4 are cells in distinct differentiation states, but we wanted to have a clearer view of the cell clusters that potentially have an essential role in the regeneration process. For this, we made another subset of cells that corresponded to C0, C1, C3 and C4, but uniquely from cells of the anlage (**Fig 9F** [↗](#)). The rationale was that cells from the mesentery are certainly at a different state from those of the anlage and thus could interfere with the pseudotime of cells from the anlage. The RNA velocity analysis using this subset strengthened our previous inferences. First, C1 seems to be the least dedifferentiated cell cluster, whose future state will be cells of C3 and part of the population of C4. Second, that C0 seems to be in a specialized state of differentiation that has a relationship to C4 (**Fig 9G** [↗](#)). This would explain the relationship of this cluster to that of the differentiating cells of C5 and C9. Interestingly, portions of C1, C0 and C4 appear to be in an advance process of differentiation based on their longer arrows compared to C3 and another portion of C4 close to C3 (**Fig 9G** [↗](#)). The Slingshot analysis of the anlage cells from C0, C1, C3 and C4 revealed that the pseudotime starts at a point of convergence that contains a portion of cells from C1 and C0. Yet, it further supports the cells from C1 as the least differentiated (**Fig 9H** [↗](#)). C1 is then followed by cells of C3, C4 and lastly C0, which for the most part seems to be at a more advanced differentiation state with a closer relationship to differentiating cells (**Fig 9I** [↗](#)).

Discussion

In this study we employed scRNA seq and HCR-FISH techniques to examine the cellular phenotypes in regenerating intestinal tissues of the sea cucumber *H. glaberrima*. These techniques have been seldom used to characterize echinoderm cells, and the few studies available are mainly limited to embryonic stages of sea urchins (*Strongylocentrotus purpuratus*, *Lytechinus variegatus*) and the sea stars (*Patiria miniata*) (Cocurullo et al., 2023 [↗](#); Foster et al., 2022 [↗](#); Meyer et al., 2023 [↗](#); Paganos et al., 2022a [↗](#), 2022b [↗](#), 2021 [↗](#); Tominaga et al., 2023 [↗](#)). Nonetheless, these studies provide an excellent description of the cell population and dynamics arising from major germ lines during echinoderm development. We now apply the same techniques to explore cell phenotypes involved in intestinal regeneration in holothurians. This is, to our knowledge, the first-time scRNA-seq and HCR-FISH have been used in adult echinoderms to analyze the cellular and molecular basis of their amazing regenerative properties. This research integrates the extensive cellular and molecular information on intestinal regeneration in holothurians collected over the past two decades, offering a comprehensive view of the cellular phenotypes and molecular changes involved.

Cell types and differentiation stages in the 9-dpe regenerating intestine

Our study has focused on the description of cells present in the 9-dpe regenerating intestine of the sea cucumber, where cells are known to have gone through dedifferentiation and are, at this time-point, in the process of differentiating into specialized cells (García-Arrarás et al., 2011 [↗](#)). Based on our analysis we have identified 13 distinct populations that form part of the regenerating

intestinal mesentery and anlage (**Fig 1B**). Among these clusters we have described the clusters corresponding to the coelomocyte, coelomic epithelium, and mesenchyme cell types (**Fig 3**). Thus, we can now use the information that our laboratory and others have gathered to pinpoint the cellular mediators and their activity during regeneration. The coelomic epithelia, for example, is highly influential in the regeneration process as it is the site where major cellular events occur, particularly cell division, dedifferentiation, and differentiation (García-Arrarás et al., 2011). Moreover, the mesenchyme ECM is known to undergo remodeling during the regeneration process, which is also critical for the proper growth of the new tissue (Quiñones et al., 2002). However, while much of the events have been described using microscopic and histological tools, we need a more comprehensive understanding of the transcriptomic characteristics of specific cell populations involved in the regeneration process.

The sea cucumber contains various mesenchymal and coelomocyte populations

Amongst the phenotypes identified are clusters of mesenchymal (C2 and C7) and coelomocyte cell populations (C6, C10, C11, and C12). The mesenchymal populations demonstrated a unique expression of *ERG* and *ETS-1*, markers of mesenchymal populations in other echinoderms. *ETS-1* has long been associated with developmental processes of mesenchymal formation in sea urchin (Koga et al., 2010; Rizzo et al., 2006). In other studies, *ERG*, a member of the ETS gene families, has been found to be necessary for controlling mesenchymal identity and differentiation (Cox et al., 2014; Mochmann et al., 2013). Further analyses of their individual marker genes suggest that these cells might be involved in EMT. For instance, C2 expresses *TIMP3*, a metalloproteinase inhibitor that aids in the extracellular matrix remodeling (Dewing et al., 2020), and *NPNT*, a gene reported to be related to development and cancer processes (Magnussen et al., 2021). These genes are also crucial for cells undergoing EMT as the cells need to detach from the other cells and the basal lamina that forms the epithelium. Comparatively, the expression profile of C7 with genes such as *PA21B* (phospholipase A2), *TMPS9* (transmembrane protease serine 9), *SEPP1*, *DMBT1*, *ITIH3*, and *SVPE1* and its GO results suggest this mesenchymal population is undergoing different processes. Based on these contrasting expression profiles, we propose that C2 corresponds to cells that have recently undergone EMT from the coelomic epithelium and eventually differentiate into a more specialized phenotype (C7). Our pseudotime results further support this developmental transition as cells in C7 appears to be in a more advanced stage when compared to those of C2 (**S6 Fig**). Therefore, these two populations correspond to the first transcriptomic description of mesenchymal phenotypes reported in sea cucumber regenerating intestine.

Our dataset contains four distinct populations that we have characterized as coelomocytes. The coelomocyte populations reported in different holothuroid species ranged between 4 to 6 distinct types (Hetzel, 1963; Ramírez-Gómez et al., 2010; Xing et al., 2008). Studies from our laboratory previously revealed four different coelomocyte populations in *H. glaberrima*, distinguished by their morphology, histochemistry, and phagocytic activity. These were lymphocytes, phagocytes, spherulocytes and a population named “giant cells” (Ramírez-Gómez et al., 2010). The distinctive gene expression profile of each of the coelomocytes that we have identified can provide insights into the differences in their role as immune/circulating cells within the sea cucumber. For example, the C6 marker gene fibrinogen-like protein, which is part of a protein family known as FREP, makes this population of great interest.

Mainly because these molecules have been vastly studied across invertebrates, and multiple immune roles have been proposed, including phagocyte recognition and encapsulation (Hanington and Zhang, 2011). A distinct example is that of C10, where the expression of *HCK*, *FER* and *ITF8* markers suggest this might be a macrophage-like activity (Chen et al., 2023; Dolgachev et al., 2018; Shuttleworth, 2018). Specifically, *HCK*, a member of Src family kinases (SFK), has been closely related to macrophage activation and polarization (Bhattacharjee et al., 2011; Poh et al.,

2015 [\[1\]](#)). Interestingly, recent studies in *A. japonicus* found that an Src homolog mediates the phagocytosis of *Vibrio splendidus*, which further supports C10 immune identity (Wan et al., 2022 [\[2\]](#)). Thus, to our knowledge, this would be the first report of the expression profile of distinct coelomocyte populations in an adult echinoderm species, setting up the stage for integrating these populations with the previously described ones.

The coelomic epithelia of the intestinal anlage is composed of a heterogenous population of cells

Our research findings align with previous microscopic descriptions of cells in the normal and regenerating mesothelium and coelomic epithelia. In our data, we most easily identify the cell population that forms the cluster exhibiting a proliferative phenotype (C8). Proliferative cells in the regenerating intestine are primarily localized within the coelomic epithelium of the anlage (García-Arrarás et al., 2011 [\[3\]](#), 1998 [\[4\]](#); Reyes-Rivera et al., 2024 [\[5\]](#)), and the gene expression profile documented here unequivocally identifies these as proliferative.

A second population of cells that can be well correlated to previously described cells is that with a muscle cell phenotype (C5). The evidence suggests that this cell population represents those cells from the coelomic epithelium that are differentiating into myocytes. This evidence includes: (1) the top-expressed genes by the cells in this cluster are all muscle-associated genes; (2) the top-enriched terms are all related to muscle morphogenesis; (3) the cluster is mostly composed of cells that come from the anlage where muscle formation is known to be taking place at this stage (Murray and García-Arrarás, 2004 [\[6\]](#)); (4) differentiated muscle cells were not dissociated by the enzymatic procedure strongly suggesting that the muscle cells that we have identified in our data are those that are in a differentiation process, rather than fully differentiated cells closer to the body wall; (5) the cells in this cluster were identified by HCR-FISH of *MYH7*. These cells are localized toward the basal region of the coelomic epithelium, the region where the differentiating myocytes are known to be present (Murray and García-Arrarás, 2004 [\[6\]](#)). Moreover, among the top expressed genes of this cluster is Troponin I (*TNNI1*), which has also been reported to be highly expressed in muscle precursor cells of the sea star embryo and immature cardiomyocytes of the chicken (Mantri et al., 2021 [\[7\]](#); Tominaga et al., 2023 [\[8\]](#)). The high differentiation activity of these cells is also supported by our pseudotime analysis, where distinct cells within the cluster are in individual differentiation states (Fig 9D [\[9\]](#) & E). Further analysis of these populations could allow us to understand the transcriptional changes these cells undergo to become fully specialized muscle cells.

An additional population in our dataset, is the neuroepithelial population (C9). This population has *STARD10*, among its top expressed genes, known to be a phospholipid transfer protein present in a neural cell population localized in the coelomic epithelium of the intestine (García-Arrarás et al., 2019 [\[10\]](#); Rosado-Olivieri et al., 2017 [\[11\]](#)). Furthermore, cells of this cluster have high expression of other genes reported to be expressed by neuronal cells of regenerative and developmental tissues, such as beta-tubulin in developing human gut and sea urchin larva, and synapsin in planaria regenerating tissue (Cocurullo et al., 2023 [\[12\]](#); Elmentaite et al., 2020 [\[13\]](#); King et al., 2024 [\[14\]](#)). Interestingly, an earlier study showing that beta-tubulin positive cells arise from dedifferentiated cells of the regenerating intestinal tissue of the sea cucumber (García-Arrarás et al., 2019 [\[10\]](#)) also suggests that similar to the muscle population, these cells are neuroepithelial cells that are differentiating rather than fully specialized. Further support for this theory lies in the main contribution to C9 coming from anlage cells and their terminal differentiation state observed in the pseudotime analysis (Fig 1D [\[15\]](#) & 10D [\[16\]](#)).

The remaining four clusters of the coelomic epithelium supra-cluster are more challenging to characterize. Nonetheless we will explore some hypotheses regarding their cellular phenotypes. The identity of C4 was put forward based on its close transcriptional and pseudotime correlation with populations of the coelomic epithelia (Figs 3 [\[17\]](#) and 9 [\[18\]](#)). Additionally, in our *in situ*

hybridization, its marker gene *SC6A5*, is mainly expressed by the coelomic epithelial cells of the normal intestine, strongly suggesting that the cells expressing this gene in the anlage are those that will become the coelomic epithelial cells of the regenerated organ (**Fig 7** [↗](#)).

C0 remains an intriguing cell population. On one hand it appears to be closely related, by its gene expression, to other coelomic epithelial populations (**Fig 3** [↗](#)). Thus, it may represent an unknown cell population or cell stage. The *in situ* expression suggest that this cluster represents the cellular population of the mesentery or peritoneal coelomic epithelium. In this case we might be evidencing differences in the coelomic epithelium of the mesentery (exemplified by cluster 0) from those of the coelomic epithelium of the intestine (exemplified by cluster 4). Future experiments will be needed to address this controversy.

Lastly, C1 and C3 share many common genes, and represent distinct populations but are still closely associated, regardless of the clustering parameters or the statistical assessment performed. Although we considered combining them into a single cluster, we ultimately decided against it, as they likely represent different stages of cell development or plasticity in the regenerating intestine. One-to-one comparisons revealed that C1 expressed various ribosomal gene markers, while C3 expressed specific genes such as *EPHA4* (ephrin type-A receptor 4) and *LRIG3*, indicating distinct transcriptomic states. Additionally, all trajectory analyses revealed that C1 cells appear to give rise to C3 cells.

Further characterization of the least differentiated cells, C1 and C3, within the anlage coelomic epithelium suggest that these cells probably serve as cellular precursors to differentiating cells. The evidence from pseudotime analyses and gene expression of C1 and C3 strongly support this conclusion. These cells appear to exhibit pluripotency with the potential to form muscle, neurons, coelomic epithelia and mesenchymal cells in the regenerating intestine. Their gene expression profiles include markers from gene families associated with embryonic development such as Hox, zinc fingers, basic helix-loop-helix and others, some of which are linked to stemness and pluripotency. For example, *HSP90A*, a molecular chaperone essential for stem cell pluripotency, and markers like *HES1* and TGFb, essential for maintaining stem cell proliferation, are present in these cell populations ([Aztekin, 2021](#) [↗](#); [Mishra et al., 2005](#) [↗](#)). Moreover, the expression of *PIWL1* (piwi-like protein 1) and *YAP1* further support the classification of these clusters as precursor cells of the intestinal anlage. In hydra, piwi-like molecules are exclusively localized in stem/progenitor cells ([Juliano et al., 2014](#) [↗](#)), and in mice, *YAP1* is expressed only in multipotent cells during intestinal epithelium regeneration, vital for their emergence ([Ayyaz et al., 2019](#) [↗](#)). Ultimately, trajectory analysis also supports C1 as a precursor cell population, as the dividing cells (C8) appear to give rise to C1, which then progresses toward C3. This analysis aligns with evidence of a proliferation center in the coelomic epithelia that provides precursor cells essential for the growth of the anlage.

C1 and C3 cells are probably dedifferentiated cells from the mesentery mesothelium, retaining markers common to all epithelial cells. Similar to what is known in other highly regenerative organisms, these dedifferentiated cells can proliferate and then re-differentiate into the cells of the new organ. Future experiments will determine whether they retain the memory of their previous phenotype (muscle or coelomic epithelium) or are completely pluripotent.

The coelomic epithelium of the intestinal anlage is pluripotent

Regardless of which cell population is responsible for giving rise to the cells of the regenerating intestine, our study reveals that the coelomic epithelium, as a tissue layer, is pluripotent. Thus, it is capable of giving rise to various cell types. Microscopy studies across different echinoderm species have consistently suggested that the coelomic epithelium can differentiate into various cell types. Among these are the formation of muscle cells and neurons in sea cucumbers ([Yu Dolmatov et al., 1996](#)), mesenchymal cells in brittle star ([Piovani et al., 2021](#) [↗](#)), and even immune cells

(coelomocytes) in sea star (Byrne et al., 2020 [↗](#); Sharlaimova et al., 2021 [↗](#)). In some species, the coelomic epithelium has even been proposed to transdifferentiate into intestinal luminal cells (Mashanov et al., 2005 [↗](#)).

The direct association of pluripotency with the coelomic epithelium is evident in both the homeostatic regenerative processes that maintain cellular numbers and in the regenerative responses to injury or autotomy. In some cases, pluripotency is directly associated with the coelomic epithelium of the regeneration anlage. This is exemplified in the brittle star *Marthasterias glacialis*, where it has been suggested that the coelomic epithelium is not only involved in arm regeneration but also serves as a source of immune cells (Guatelli et al., 2022 [↗](#)). This report indicated that “residential stem cells” in the regenerating arm of the brittle star originate from the coelomic epithelia (Candia-Carnevali et al., 2009 [↗](#)). Similarly, in the sea cucumber *Holothuria forskali*, the injured mesothelial layer has been identified as the source of undifferentiated cells that differentiate into the cells of the growing organ and into phagocytic cells, now recognized as coelomocytes (Vandenspiegel et al., 2000 [↗](#)).

Our scRNA-seq provides convincing evidence that various cell populations originate from the coelomic epithelia of the sea cucumber anlage, offering a more detailed view of the differentiation process. These cell populations, now classified by their gene expression data, correlate well with those previously characterized by microscopy descriptions. We have strong evidence for muscular, proliferating and neuroepithelial cells (C5, C8, and C9) arising from the coelomic epithelia of the anlage. We also show evidence of a differentiating cell population corresponding to the mature intestine coelomic epithelium (C4). Furthermore, our data suggest the presence of a mesenchymal population arising from the coelomic epithelium of the anlage through EMT (García-Arrarás et al., 2011 [↗](#)).

Of particular interest, our study reveals that the mesentery coelomic epithelia population (C0) exhibits localized expression of *SAA1*, an immune response related gene. This finding, in line with previous reports of *SAA1* localization in the coelomic epithelium of the regenerating intestinal tissue (Santiago et al., 2000 [↗](#)) (**S5 Fig** [↗](#)), raises the intriguing possibility that the coelomic epithelium of the sea cucumber regenerating intestine could also be the source of coelomocytes, as suggested by Guatelli (2022) (Guatelli et al., 2022 [↗](#)). This hypothesis is further supported by the observation that in the sea star *Asteria rubens*, coelomocytes arise from its coelomic epithelium (Vanden Bossche and Jangoux, 19976). However, our current results, while suggestive, do not yet provide conclusive evidence to support this in the sea cucumber, at least not in the coelomic epithelia of the regenerating intestine or the 9-dpe stage.

Currently, there is no definitive evidence identifying the precursor cells that give rise to these cell populations in echinoderms. However, given our gene expression data and the interactions observed among the cell populations, we postulate that cells from C1 stand as the precursor cell population from which the rest of the cells in the coelomic epithelium arise. Granted, much more experimental evidence will be needed to arrive at this conclusion.

Nonetheless, it provides a focus on the most likely, and probably most intriguing cell population. It also provides the path to explore multiple questions that arise from our results. Can C1 be subdivided into other cell populations? Are the cells in this cluster pluripotent? Do these cells originate from the same cells via dedifferentiation? These and many other questions will indeed be tested in future experiments.

Our data has allowed us to construct a model of the cell populations we identified in the 9-dpe intestinal anlage (**Fig 10** [↗](#)). This model presents the coelomic epithelium of the anlage as a heterogeneous layer of cells comprising of six cell populations corresponding to distinct differentiation states. Among these are cell populations in the process of differentiation: muscle, neuroepithelium, and coelomic epithelium cells. We also propose the presence of undifferentiated

and proliferating cell populations in the coelomic epithelia, which give rise to the cells in this layer. Underneath the coelomic epithelia are the mesenchymal cells, some of which may have originated from the coelomic epithelial layer via EMT. Not shown in our model is a population localized in the coelomic epithelium of the mesentery that we infer also plays an important role in the regeneration process of the intestine.

The intestinal anlage as a regeneration blastema

The intestinal anlage is a particular regenerative structure that has long been described as a blastema-like structure due to its remarkable morphological resemblance to the “classical blastema” (García-Arrarás et al., 1998 [\[1\]](#)). As stated previously a blastema is usually described as a transient structure composed of a mass of proliferating undifferentiated cells. This structure is found at the site of injury and will give rise to the regenerated organ (Seifert and Muneoka, 2018 [\[2\]](#)). The blastema cells can originate from different lineages across species. For example, in amphibians, the blastemal cells originate from dedifferentiated muscle, cartilage, fibroblast and other tissues, while in the flatworm they originate from undifferentiated stem cells known as neoblasts (Baguñà, 2012 [\[3\]](#); Globus et al., 1980 [\[4\]](#); Wagner et al., 2011 [\[5\]](#)). The blastema is overlaid by a wound epidermis that is formed by a re-epithelization process following injury, which develops into a specialized wound epidermis that in amphibians is known as the apical epithelial cap (AEC) (Aztekin, 2021 [\[6\]](#)). Since its first description, more than 100 years ago, the blastema has been regarded as the best indicator of regeneration, and its presence is associated with tissues or organs that are highly regenerative. In fact, it has been proposed that the presence or absence of a blastema defines the regenerative success or failure of the regeneration process. However, as more regenerative species continue to be studied, the classical definition of a blastema has been reconsidered, moving towards its functional role rather than its histological or structural characteristics (Seifert and Muneoka, 2018 [\[2\]](#)).

The blastema of salamanders and newts, have long served as models to describe a blastema cellular and molecular properties (Globus et al., 1980 [\[4\]](#); Scimone et al., 2022 [\[7\]](#); Tajer et al., 2023 [\[8\]](#)). The sea cucumber intestinal anlage stands as a different structure in terms of the tissue compartmentalization of certain activities. Yet, when examined closely, much of the processes and signaling molecules described in the amphibian blastema also take place in the holothurian anlage, although their spatial occurrence might differ. For instance, like the blastema cells of the amphibians, the cells of the holothurian anlage coelomic epithelium are proliferative undifferentiated cells and originated via a dedifferentiation process. Moreover, some of the genes expressed by the amphibian blastema cells or the overlying AEC, such as Wnt, Tgf-beta, and Fgf, are also known to be expressed during sea cucumber regeneration (Auger et al., 2023 [\[9\]](#); Zeng et al., 2023 [\[10\]](#)). In amphibians, some of these factors are thought to be released by cells of the AEC, serving as a way of modulating the blastema cell activity. For example, FGF1 is expressed in the cells of the AEC while FGF receptors were localized to blastema cells (Zenjari et al., 1996 [\[11\]](#)). Moreover, both FGF and Wnt signaling have been shown to be vital for the formation of the blastema (Makanae et al., 2014 [\[12\]](#)).

This expression of Fgf and Wnt and their possible functions correlates with what has been found in sea cucumbers. Not only are the same growth factors expressed in C1 and C3, but a recent study highlighted the role of *FGF4* as a modulator of cell proliferation during the intestinal regeneration of the sea cucumber *A. japonicus* (Zeng et al., 2023 [\[10\]](#)). This group demonstrated that inhibiting *FGF4* and its receptor, *FGFR2*, negatively affected cell proliferation of the mesothelial layer during intestinal regeneration. Similar results have been shown for Wnt, primarily a study by our laboratory demonstrating that Wnt pathway inhibition, either by pharmacological drugs or by RNAi caused a reduction in cell proliferation (Alicea-Delgado and García-Arrarás, 2021 [\[13\]](#); Bello et al., 2020 [\[14\]](#)). Thus, in the sea cucumber coelomic epithelia of the anlage, cells are also involved in intercellular communications that modulate cellular dynamics through factors similar to those found in the AEC-blastema cell modulation.

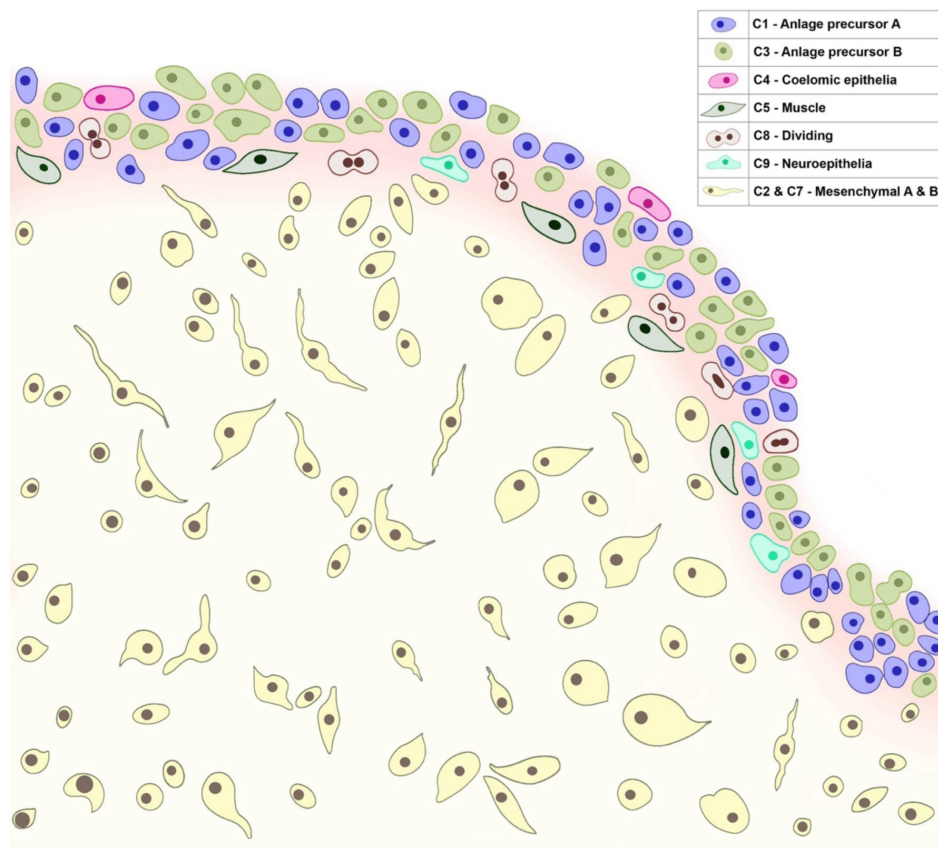


Fig 10.

Model of cellular organization within the anlage of the regenerating intestine of the sea cucumber *H. glaberrima* at 9 dpe.

Schematic highlights the heterogeneity of the coelomic epithelia of the intestinal anlage and identified mesenchymal populations. The colors of cells and C# correlate with those used on the UMAP of [Fig 1B](#).

The holothurian anlage and the amphibian blastema exhibit some differences. In the latter there is a clear separation of the AEC and the underlying blastema cells. In amphibians, the AEC has been shown to actively participate in the formation and maintenance of the blastemal cells, particularly in their proliferation and differentiation. In contrast, in holothurians, cells with both blastema and AEC characteristics can be found within the coelomic epithelium. In fact, both *PRRX1* (paired mesoderm homeobox protein), a gene expressed by amphibian blastema cell precursors (Gerber et al., 2018 [↗](#); Lin et al., 2021 [↗](#)), and *HES1* (transcription factor HES-1), a marker gene of the amphibian AEC, are expressed by the holothurian coelomic epithelium (Aztekin, 2021 [↗](#)) (**Fig 3** [↗](#) and **S5 Fig** [↗](#)).

In conclusion, in all cases where it has been studied, the blastema appears as a dynamic structure where communication among its cells and the overlying epidermis or epithelium plays important roles in the regenerative process. In this context, the sea cucumber intestinal anlage could be considered a blastema since it uses similar mechanisms to fulfill the same role in the regeneration of the new organ. In this case, the cells of the coelomic epithelium would be those considered to be the blastemal cells.

Methods

Animals and Sample Collection

Two adult sea cucumbers were collected from the northeastern rocky region of Puerto Rico and were kept in aerated sea water for acclimatation prior to experiments. Evisceration of the sea cucumber was stimulated by intracoelomic injection of 0.35 M KCl as previously described (Reyes-Rivera et al., 2024 [↗](#)). They were maintained in sea water aquaria for 9 days to undergo regeneration until dissection. Before the dissection, animals were anesthetized by immersion in ice-cold water for 45 minutes. Dissection was done through an initial dorsal incision which allowed exposition of the internal organs, upon which the growing rudiment (or anlage) and mesentery were dissected and separated from each other. Each tissue was kept in CMFSS on ice and treated separately during the tissue dissociation process.

Tissue Dissociation

Mesentery and rudiment tissues were digested for 15 minutes, in rocking shaker at room temperature with 1mL of 0.05% Trypsin/0.02% EDTA solution prepared in CMFSS (Bello et al., 2015 [↗](#)). Digestions were quenched by adding 500μL of 0.2% BSA in CMFSS and then centrifuged for 2 minutes at 1500 rpm. The supernatants were discarded, and the pellets resuspended with 500μL of 0.04% BSA in CMFSS and the cells were gently separated by pipetting using glass pipets with fire blunted tips. The cell suspensions were filtered using a nylon cell strainer with 70mm mesh and aliquots were taken for cell counting. Cell viability was assessed through manual cell counting with hemocytometer to confirm that a viability higher than 90% was maintained. Sample suspensions were adjusted to 1000 cells/mL using CMFSS, as required for sequencing procedures.

Immuno- and cytochemistry

Fifty μL of dissociated cell sample were placed on polysine-treated slides and fixed with 50 μL of 4% paraformaldehyde and left to dry overnight. Slides were washed in PBS prior to use for immune and or cytochemistry. The methodology for the immunofluorescence techniques/preparation of slides was followed as published except for the time of PBS washes which were of 10 minutes instead of 15 minutes (Diaz-Balzac et al., 2007 [↗](#); Reyes-Rivera et al., 2024 [↗](#)). The primary and secondary antibodies used are in **S7 Table** [↗](#). Samples to be probed with fluorescent phalloidin were treated directly with Phalloidin-TRITC (1:1000; Sigma P1951) for

1h as described previously (Reyes-Rivera et al., 2024 [DOI](#)). DAPI was incorporated into the mounting medium as described previously (Reyes-Rivera et al., 2024 [DOI](#)). Cells were observed in a Nikon DS-Qi2 fluorescent microscope.

Single Cell RNA Sequencing and Data Analysis

Single cell libraries were prepared using the 10X Genomics Chromium Next GEM Single Cell 3' Kit v3.1 and the Chromium 10X instrument, following the protocol from manufacturer (CG000204). Sequencing was carried out with Illumina NextSeq 2000 at the Sequencing and Genotyping Facility of the University of Puerto Rico Molecular Science Building.

The raw sequencing reads from all four samples (two mesentery and two anlage) were processed individually using Cell Ranger (v7.1.0) (Zheng et al., 2017 [DOI](#)) using the reference genome and gene models of *H. glaberrima* available at blastkit.hpcf.upr.edu/hglaberrima-v1. Gene models were annotated against human reference proteins from Uniprot and the entire Uniprot reference protein database. Gene IDs throughout the study correspond to the annotations with the human reference protein sequences from Uniprot.

Initial quality control was conducted in R (v4.3.2) using SoupX (v1.6.2) with default parameters to remove ambient RNA (Young and Behjati, 2020 [DOI](#)). Subsequent data processing was carried out using Seurat (v4.3.0.1) (Hao et al., 2024 [DOI](#), 2021 [DOI](#); Satija et al., 2015 [DOI](#); Stuart et al., 2019 [DOI](#)). After data normalization and identifying variable features (n=2000), 22,970 anchors were identified using 20 dimensions. The datasets were integrated using the canonical correlation analysis (CCA) approach of Seurat, followed by data scaling for principal component analysis (PCA) (npcs = 50) and Uniform Manifold Approximation and Projection (UMAP) dimensional reduction technique (dims = 1:20).

Neighbors were then identified with 20 dimensions, leading to cluster identification at a resolution of 0.5. Various resolution parameters (0.8, 1.2, and 1.4) and dimensions (up to 35 in increments of 5) were tested before the final resolution value. The statistical analysis tool scSHC was employed to assess the probability of each cluster being unique (Grabski et al., 2023 [DOI](#)). Marker genes of each cluster or supra-clusters were identified using the FindMarkers() function in Seurat and mapped to the appropriate Uniprot IDs. Genes highlighted throughout the study were further validated using the NCBI non-redundant reference database and EchinoBase (Telmer et al., 2024 [DOI](#)) via BLAST.

Differential expression data across clusters was used to perform gene set enrichment analysis of Gene Ontology (gseGO) biological processes terms using clusterProfiler (v10.0) (Yu et al., 2012 [DOI](#)) with a p-value cutoff of 0.05. For this analysis, BLASTp was used to map *H. glaberrima* gene models to human reference protein sequences from NCBI (GCF_000001405.40), facilitating the assignment of ENTREZ ID to the correct human GO terms in the AnnotationDbi (v1.64.1) human database (v2.1) (Pagès et al., 2024 [DOI](#)).

RNA velocity loom files were generated with velocity (v0.17), which relied on genome-masked regions obtained from RepeatModeler (v2.0.5) and RepeatMasker (v4.1.5) (Smit et al., 2015 [DOI](#); Smit and Hubley, 2015 [DOI](#)), along with the sea cucumber gene models and CellRanger dataset. These files were further analyzed with velocity.R (v0.6) and SeuratWrappers (v0.2.0), using the ReadVelocity and RunVelocity functions. Pseudotime analysis for the distinct data subsets was conducted using Slingshot (v2.10.0) (Street et al., 2018 [DOI](#)). The code of the data analysis has been made available at GitHub (https://github.com/devneurolab/scRNAseq_Hglaberrima [DOI](#)).

HCR-FISH

Regenerating intestines from animals eviscerated 8-9 days previously and intestines from non-eviscerated (controls) animals were collected and fixed in 4% (v/v) paraformaldehyde with phosphate-buffered saline (0.01M PBS; 0.138M NaCl; 0.0027M KCl; pH7.4) overnight at 4 °C. Tissues were then, washed three times by PBS, and treated overnight with 40% saccharose prior to cutting in the cryostat. Sections (20 µm) were prepared in a cryostat (Leica CM1850), as previously published for immunohistochemistry (Reyes-Rivera et al., 2024 [DOI](#)). Gene spatial expression was determined by designing twelve (12) or twenty (24) set split probes for each gene marker. To decrease probe unspecific binding and increase its signal to background ratio, probes were designed as described by H. Choi and colleagues (Molecular Instruments) (Choi et al., 2018 [DOI](#)). To ensure probe gene target specificity, all nucleotide regions selected for probe design underwent extensive validation using the newly developed *H. glaberrima* genome and transcriptome alignment tool (<https://blastkit.hpcf.upr.edu/hglaberrima-v1/> [DOI](#)) (Medina-Feliciano et al., 2021 [DOI](#)).

Probes were used at a final concentration of approximately 20-25 nM. Immediately after slide preparation, HCR-FISH v3 was carried out using a modified version of Molecular Instrument's fresh fixed frozen tissues protocol. Probe hybridization was conducted overnight at 37 °C, while DNA hairpin amplification (B1-546nm or B2-546nm) was done at 25°C overnight with 3pmol of h1 and h2, respectively. Our protocol modification involved replacing Ethanol with Methanol during sample permeabilization. Also, the use of proteinase K was omitted, and Tween 20 at 0.1% was added to all PBS wash buffers. Once slides were prepared, image acquisition was attained using a Nikon DS-Qi2 fluorescent microscope. Positive control probes (PolyA) were used as signal calibrators to define background from positive signal during image analysis.

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Data Availability

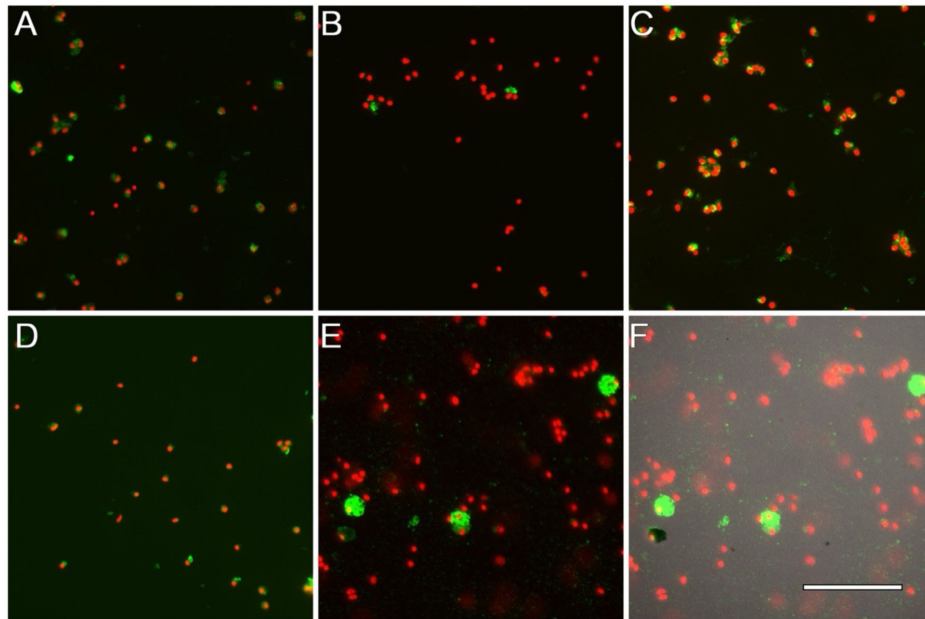
Data generated during this project has been made publicly available at Figshare (Medina-Feliciano et al., 2024 [DOI](#)) including raw and processed sequencing data.

Supplement

S1 Fig.

Labeling of dissociated cell phenotypes with cell markers.

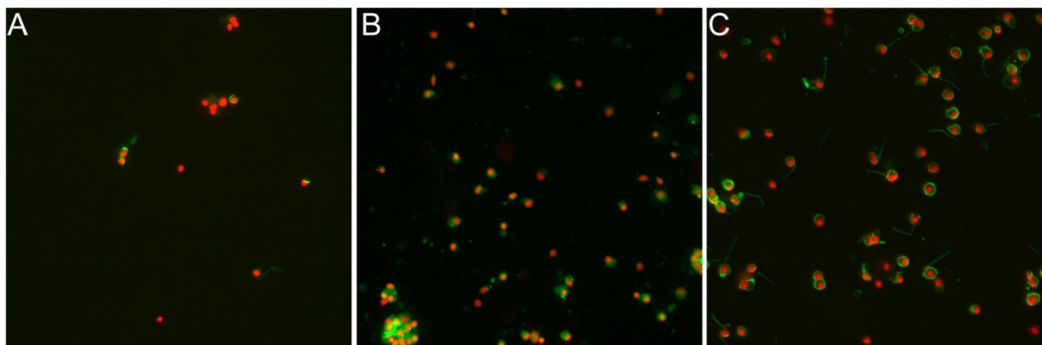
Immunocytochemistry, fluorescently labeled phalloidin and Toluidene blue were used to identify various cell populations among the cells dissociated from the mesentery an anlage sample. These include A. a mesenchymal marker (KL4), B. a neuronal marker (RN1), C. a mesothelial marker (Meso1), D. a muscle marker (Phalloidin) and E & F two coelomocyte markers (The antibody SphAA12 and toluidine blue). E. Immunocytochemistry using the SphAA12 antibody F. Overlay of toluidene blue labeling (see dark cell on lower left) identifies a different coelomocyte population.



S2 Fig.

Immunocytochemical labeling of dissociated cell phenotypes using three different antibodies against tubulin.

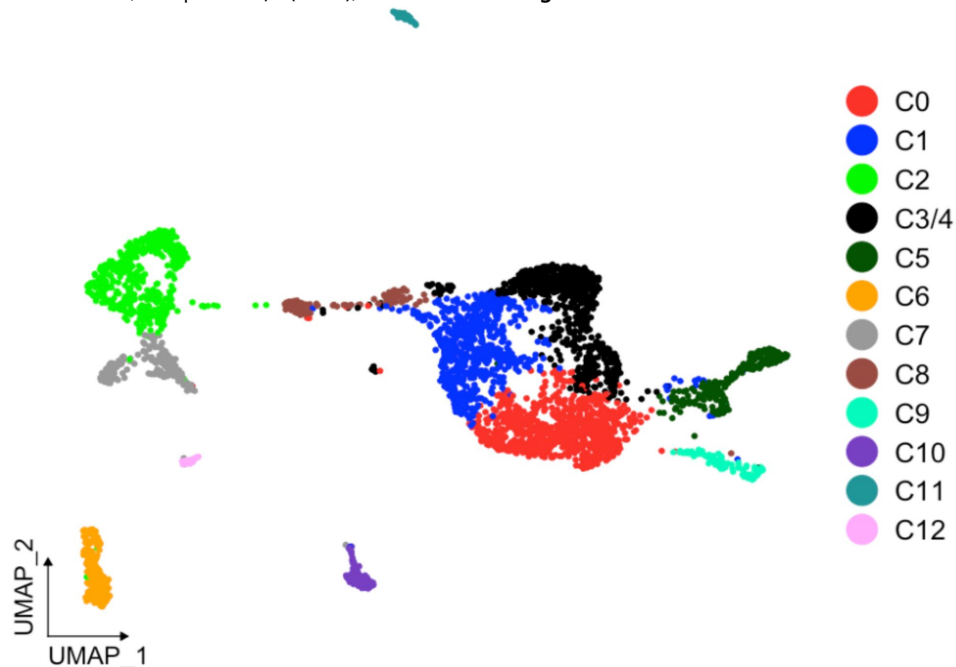
A. anti-acetylated tubulin labels around 7% of the cells. B. Anti-beta-tubulin labels around 70% of the cells. C. Anti-alpha tubulin labels around 80% of the cells.



S3 Fig.

UMAP of clusters after statistical assessment with scSHC.

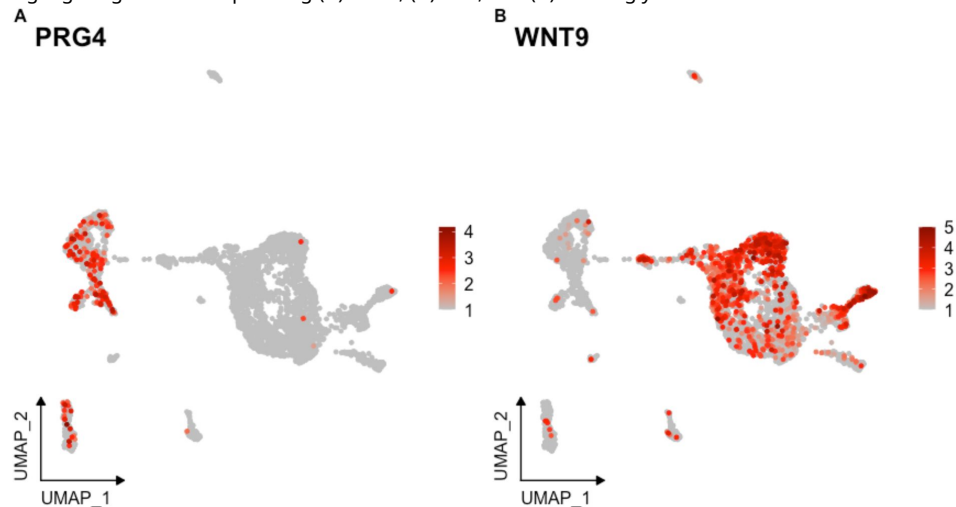
Results reflect that each of the identified clusters are unique, except for C3 and C4, which it suggests they correspond to a single cluster. All clusters, except for C3/4 (black), are colored as in **Fig 1B**.



S4 Fig.

Expression of marker genes previously documented in the sea cucumber.

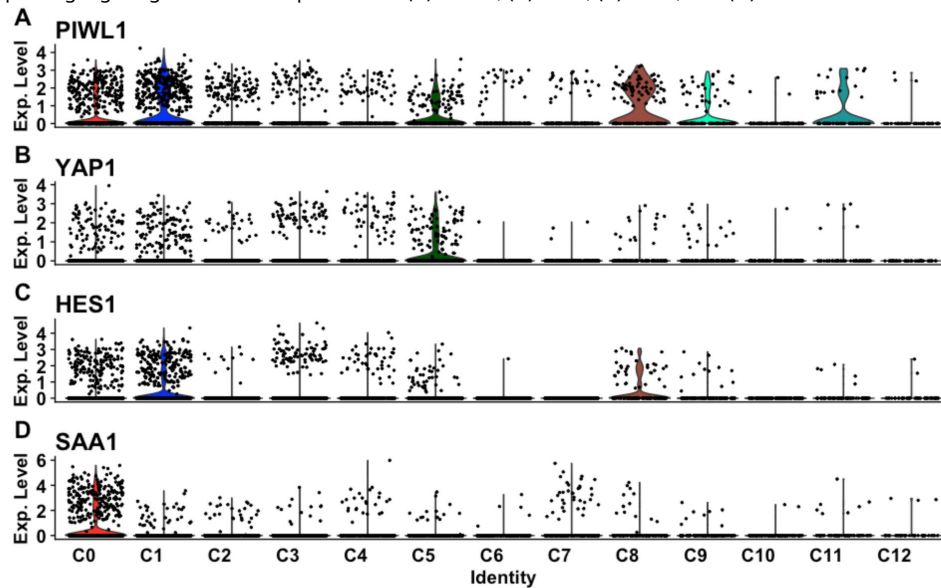
(A-C) UMAP highlighting the cells expressing (A) Wnt9, (B) SAA, and (C) Proteoglycan-4.



S5 Fig.

Expression of marker genes associated with cell types or state.

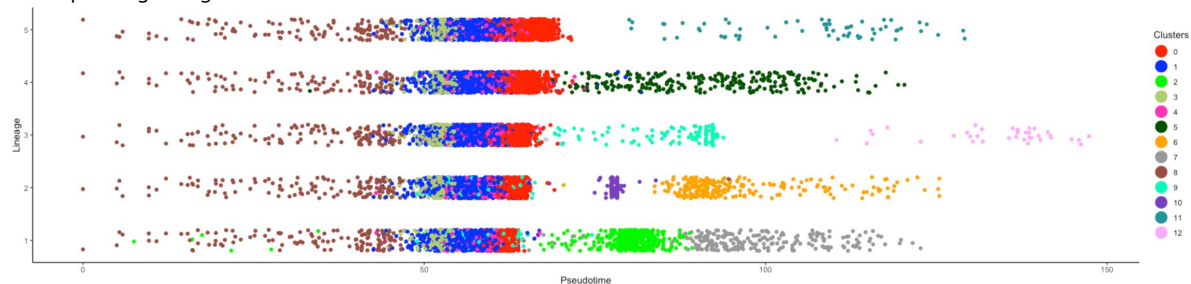
(A-D) Violin plot highlighting the level of expression of (A) PIWL1, (B) YAP1, (C) HES1, and (D) SAA1 across clusters.



S6 Fig.

Pseudotime analysis of all clusters within the 9dpe intestinal regeneration dataset.

Each color reflects the corresponding cluster from C0-C12. The x-axis represents slingshot pseudotime and y-axis the corresponding lineage.



Antibody/Marker	Labeling	Product No.	Reference
Ab Sph AA12	Coelomocyte	-	Ramírez-Gómez et al., 2010
Ab RN1	Neuronal	-	Nieves-Ríos et al., 2020
Ab GFS	Neuronal	-	Díaz-Balzac et al., 2007
Ab KL14	Mesenchyme	-	In preparation
Ab MESO	Mesothelial cells	-	García-Arrarás et al., 2011
Ab beta tubulin	Neuronal/Cilia	Sigma T5293	Tossas et al., 2014
Ab Acetylated tubulin	Cellular	Sigma T6793	Tossas et al., 2014
Ab alpha tubulin	Cellular	Sigma T5168	Tossas et al., 2014
Fluorescent Phalloidin	Muscle fibers	Sigma P1951	García-Arrarás et al., 2011; Murray and García-Arrarás et al., 2004
Toluidine Blue	Nuclei		

S7 Table.

Antibody/Markers used for immune- and cytochemical labeling of dissociated cell suspension.

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Reviewer #1 (Public Review):**Summary:**

Medina-Feliciano et al. investigated the single cell transcriptomic profile of holoturian regenerating intestine following evisceration, a process used to expel their viscera in response to predation. Using single cell RNA-sequencing and standard analysis such as "Find cluster markers", "Enrichment analysis of Gene Ontology" and "RNA velocity", they identify 13 cell clusters and potential identity. Based merely on bioinformatic analysis they identified potentially proliferating clusters and potential trajectories of cell differentiation. This manuscript represents a useful dataset that can provide candidate cell types and cell markers for more in-depth functional analysis for gaining a better understanding of the holoturian intestine regeneration. The conclusions of this paper are supported only by bioinformatic analyses, since the in vivo validation through HCR does not sufficiently support them.

Strengths:

- The Authors are providing a single cell dataset obtained from sea cucumber regenerating their intestine. This represents a first fundamental step to an unbiased approach to better understand this regeneration process and the cellular dynamics taking part in it.
- The Authors run all the standard analyses providing the reader with a well digested set of information about cell clusters, potential cell types, potential functions and potential cell differentiation trajectories.

Weaknesses:

- The entire study is based on only 2 adult animals, that were used for both the single cell dataset and the HCR. Additionally, the animals were caught from the ocean preventing information about their age or their life history. This makes the n extremely small and reduces the confidence of the conclusions.
- All the fluorescent pictures present in this manuscript present red nuclei and green signals being not color-blind friendly. Additionally, many of the images lack sufficient quality to determine if the signal is real. Additional images of a control animal (not eviscerated) and of a negative control would help data interpretation. Finally, in many occasions a zoomed out image would help the reader to provide context and have a better understanding of where the signal is localized.
- The Authors frequently report the percentage of cells with a specific feature (either labelled or expressing a certain gene or belonging to a certain cluster). This number can be misleading since that is calculated after cell dissociation and additional procedures (such as staining or sequencing and dataset cleanup) that can heavily bias the ratio between cell types. Similarly, the Authors cannot compare cell percentage between anlage and mesentery samples since that can be affected by technical aspects related to cell dissociation, tissue composition and sequencing depth.
- The Authors decided to validate only a few clusters and in many cases there are no positive controls (such as specific localization, specific function, changes between control and regenerating animals, co-stain) that could actually validate the cluster identity and the specificity of the selected marker. There is no validation of the trajectory analysis and there is no validation of the proliferating cluster with H3P or BrdU stainings.
- It is not clear what is already known about holothurian intestine regeneration and what are the new findings in this manuscript. The Authors reference several papers throughout the whole result sectioning mentioning how the steps of regeneration, the proliferating cells, some of the markers and some of the cell composition of mesenteries and anlages was already known.

<https://doi.org/10.7554/eLife.100796.1.sa3>

Reviewer #2 (Public Review):**Summary:**

This research offers a comprehensive analysis of the regenerative process in sea cucumbers and builds upon decades of previous research. The approach involves a detailed examination using single-cell sequencing, making it a crucial reference paper while shedding new light on regeneration in this organism.

Strengths:

Detailed analysis of single-cell sequencing data and high-quality RNA localization images provide significant new insights into regeneration in sea cucumbers and, more broadly, in animals.

Weaknesses:

The spatial context of the RNA localization images is not well represented, making it difficult to understand how the schematic model was generated from the data. In addition, multiple strong statements in the conclusion should be better justified and connected to the data provided.

<https://doi.org/10.7554/eLife.100796.1.sa2>

Reviewer #3 (Public Review):**Summary:**

The authors have done a good job of creating a "resource" paper for the study of gut regeneration in sea cucumbers. They present a single-cell RNAseq atlas for the reconstitution of *Holothuria glaberrima* gut following self-evisceration in response to a potassium chloride injection. The authors provide data characterizing cellular populations and precursors of the regenerating anlage at 9 days post evisceration. As a "Tools and Resources" contribution to eLife, this work, with some revisions, could be appropriate. It will be impactful in the fields of regeneration, particularly in invertebrates, but also in comparative studies in other species, including evolutionary studies. Some of these comparative studies could extend to vertebrates and could therefore impact regenerative medicine in the future.

Strengths:

- Novel and useful information for a model organism and question for which this type of data has not yet been reported
- Single-cell gene expression data will be valuable for developing testable hypotheses in the future
- Marker genes for cell types provided to the field
- Interesting predictions about possible lineage relationships between cells during sea cucumber gut regeneration

Weaknesses:

- Possible theoretical advances regarding lineage trajectories of cells during sea cucumber gut regeneration, but the claims that can be made with this data alone are still predictive
- Better microscopy is needed for many figures to be convincing
- Some minor additions to the figures will help readers understand the data more clearly

<https://doi.org/10.7554/eLife.100796.1.sa1>

Author response:

Reviewer #1

- The entire study is based on only 2 adult animals, that were used for both the single cell dataset and the HCR. Additionally, the animals were caught from the ocean preventing information about their age or their life history. This makes the n extremely small and reduces the confidence of the conclusions.

This statement is incorrect. While the scRNAseq was indeed performed in two animals (n=2), the HCR-FISH was performed in 3-5 animals (depending on the probe used). These were different animals from those used for the scRNAseq. We are partly responsible for this confusion, since we did not state the number of animals used for the HSC-FISH in the manuscript.

- All the fluorescent pictures present in this manuscript present red nuclei and green signals being not color-blind friendly. Additionally, many of the images lack sufficient quality to determine if the signal is real. Additional images of a control animal (not eviscerated) and of a negative control would help data interpretation. Finally, in many occasions a zoomed out image would help the reader to provide context and have a better understanding of where the signal is localized.

Fluorescent photos will be changed to color-blind friendly colors.

Diagrams, arrows and new photos will be included as to guide readers to the signal

or labeling in cells. In the original manuscript 6 out of 7 cluster validations included a photo of a normal, non-eviscerated control. We will make certain that this is highlighted in the resubmission and that ALL figures with HCR-FISH labeling will include data from control animals.

- The Authors frequently report the percentage of cells with a specific feature (either labelled or expressing a certain gene or belonging to a certain cluster). This number can be misleading since that is calculated after cell dissociation and additional procedures (such as staining or sequencing and dataset cleanup) that can heavily bias the ratio between cell types. Similarly, the Authors cannot compare cell percentage between anlage and mesentery samples since that can be affected by technical aspects related to cell dissociation, tissue composition and sequencing depth.

The Reviewer has correctly identified the limitations of using cell percentages in scRNA-seq analyses. However, these percentages do offer a general overview of the sequenced cell populations and highlight potential differences between samples. In addition, these percentages, as addressed by the Reviewer, not only emphasize the shortcomings of the dissociation methods but at the same time provide some explanation for the absence of particular cell populations, as we describe in the manuscript. In our future resubmission, we will acknowledge these limitations and inform readers of any potential biases introduced by relying on these numbers.

- The Authors decided to validate only a few clusters and in many cases there are no positive controls (such as specific localization, specific function, changes between control and regenerating animals, co-stain) that could actually validate the cluster identity and the specificity of the selected marker. There is no validation of the trajectory analysis and there is no validation of the proliferating cluster with H3P or BrdU stainings.

We validated the seven clusters that were important to reach our conclusions. Six of these had controls of normal (uneviscerated) intestine. Nonetheless we will increase the number of

cluster validations and include the dividing cell cluster using BrdU.

- It is not clear what is already known about holothurian intestine regeneration and what are the new findings in this manuscript. The Authors reference several papers throughout the whole result sectioning mentioning how the steps of regeneration, the proliferating cells, some of the markers and some of the cell composition of mesenteries and anlagen was already known.

The manuscript presents several novel findings on holothurian intestine regeneration, including:

- The integration of multiple cellular processes, reported for the first time within a single species, along with the identification of the specific mRNAs expressed by each involved cell population.
- A comparative analysis of the sea cucumber anlage structure, highlighting its similarities to previously described blastemal structures.
- The identification of the potential dedifferentiated cell populations that form the foundation of the anlage, serving as the epicenter for proliferating and differentiating cells.

We will ensure that these and other significant findings are prominently emphasized in the resubmitted manuscript.

Reviewer #2

- The spatial context of the RNA localization images is not well represented, making it difficult to understand how the schematic model was generated from the data. In addition, multiple strong statements in the conclusion should be better justified and connected to the data provided.

As explained above we will make an effort to provide a better understanding of the cellular/tissue localization of the labeled cells. Similarly, we will revise the conclusions so that the statements made are well justified.

Reviewer #3

- Possible theoretical advances regarding lineage trajectories of cells during sea cucumber gut regeneration, but the claims that can be made with this data alone are still predictive.

We are conscious that the results from these lineage trajectories are still predictive and will emphasize this in the text. Nonetheless, they are important part of our analyses that provide the theoretical basis for future experiments.

- Better microscopy is needed for many figures to be convincing. Some minor additions to the figures will help readers understand the data more clearly.

As explained above we will make an effort to provide a better understanding of the cellular/tissue localization of the labeled cells. Similarly, we will revise the conclusions so that the statements made are well justified.

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